HAUSDORFF DIMENSION OF THE SINGULAR SET IN OPTIMAL TRANSPORT

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Abstract. We study optimal transport between bounded smooth domains in $\mathbb{R}^n$, with smooth positive densities bounded above and below. Without convexity assumptions on the domains, we prove that the optimal map is smooth outside a relatively closed set of Hausdorff dimension at most $n-1$.

1. Introduction

Let $\Omega, \Omega^* \subset \mathbb{R}^n$, $n \geq 2$, be bounded open sets. Suppose f, g are probability densities concentrated on Ω, Ω^* respectively. Let λ, Λ be their lower and upper bounds

$$0 < \lambda \leq f, g \leq \Lambda < \infty, \text{ in their respective domains.} \tag{1.1}$$

Optimal transport problem asks for finding the cheapest way to transport mass from Ω to Ω^*, namely solving the following minimization problem:

$$\inf_{T_\sharp(f dx)=g dy} \int_\Omega \frac{1}{2}|x - Tx|^2 f(x) dx,$$

where $T_\sharp f dx = g dy$ means that $\int_{T^{-1}A} f = \int_A g$ for any Borel set $A \subset \Omega^*$. By Brenier's theorem [3, 4], there exists a unique optimal map T. Moreover, $T = Du$ for some convex function u. Hence,

$$(Du)_\#(f dx) = g dy. \tag{1.2}$$

We always use the globally Lipschitz representative of u described in Section 2.

The regularity theory for convex targets was developed by Caffarelli [6, 7, 8]. For nonconvex targets the optimal map can be discontinuous even for smooth positive densities. Yu [24] studied the nondifferentiability set of planar transport potentials with a convex source and a nonconvex target. Under mild additional conditions, away from a countable set it is a disjoint union of countably many C^1 curves. Figalli [14] proved partial regularity for general nonconvex planar domains. Figalli and Kim [16] established partial regularity in every dimension without convexity of the supports: $\{x \in \Omega : \partial u(x) \cap \Omega^* \neq \varnothing\}$ is an open set with full measure, and u is smooth and strictly convex there when the densities are smooth. De Philippis and Figalli [13] extended this to general costs. A very natural open problem was raised in [12]: Can we obtain a finer estimate for the size of the singular set?

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The singularities of convex functions have been extensively studied. We recall two results particularly relevant to the present work. Zajíček [25] proved that the nondifferentiability set of a convex function has Hausdorff dimension at most $n-1$. Mooney [22] proved that an Alexandrov solution of $\det D^2 u \geq 1$ is strictly convex outside a set of zero $\mathcal{H}^{n-1}$-measure.

For optimal transport problem, under additional geometric assumptions, finer descriptions are known: see Kitagawa and McCann [18] for separated target components, Chen and Liu [9] for two disjoint convex targets, and Chen, Li and Liu [10] for nonconvex polygonal targets in the plane, and Chen et al. [11] for nonconvex polyhedral targets in arbitrary dimensions. In the current paper, $\partial\Omega^*$ is an arbitrary smooth hypersurface.

Let

$$\Omega' = \{x \in \Omega : \exists\, r > 0,\ B_r(x) \subset \Omega,\ u \in C^\infty(B_r(x)),\ D^2 u > 0 \text{ on } B_r(x)\}$$

be the *regular set*. The *singular set* is $\Sigma = \Omega \setminus \Omega'$. The main result of this paper is the following Hausdorff dimension estimate of the singular set.

Theorem 1.1. *Let $\Omega, \Omega^* \subset \mathbb{R}^n$ be bounded connected open sets with C^∞ boundary, $n \geq 2$. Suppose $f \in C^\infty(\Omega)$ and $g \in C^\infty(\Omega^*)$ are probability densities satisfying* (1.1)*. Then the potential u in* (1.2) *satisfies*

$$\dim_{\mathrm{H}} \Sigma \leq n-1.$$

The exponent $n-1$ is sharp, as a symmetric version of Caffarelli's smooth dumbbell construction [8] yields an $(n-1)$-dimensional set of non-differentiability points for connected smooth domains and constant densities.

Let

$$A = \{x \in \Omega : u \text{ is not differentiable at } x\} \quad \text{and} \quad E = \{x \in \Omega \setminus A : Du(x) \in \partial\Omega^*\}. \tag{1.3}$$

By the regularity theory of Figalli and Kim (Lemma 2.1), $\Sigma = A \cup E$, and $\dim_{\mathrm{H}} A \leq n-1$ by Zajíček [25, Theorem 1]. Hence Theorem 1.1 is equivalent to the following statement, which is the key of the paper.

Proposition 1.2. *Under the hypotheses of Theorem 1.1,*

$$\dim_{\mathrm{H}} E \leq n-1.$$

We outline the proof of Proposition 1.2. For $0 < \alpha < 1$, a density bound for $\mu_{u+|\cdot|^2/2}$ yields

$$c|x'|^{2/\alpha} \leq u(x+x') - u(x) - Du(x)\cdot x' \leq C|x'|^{1+\alpha}$$

outside a set of zero $\mathcal{H}^{n-1+\alpha}$-measure. These estimates allow us to localize a fixed dual potential v and obtain uniform control of its sections centered at $Du(x)$, for x in a suitable countable family of subsets of E. If a section of v of height h has semiaxes $a_1 \geq \cdots \geq a_n$, the associated source ellipsoid has semiaxes h/a_i and covering cost

$$C_{\mathrm{cov}} = \frac{h^{n-1}}{a_2 \cdots a_n}.$$

Comparing this cost with finite weighted measures separates the positive upper mass limit from the zero mass limit. A selection argument treats the former. In the latter, two volume estimates in dimension two, and an additional measure on the subgradient graph in higher dimensions, yield a rank-one limit. The limiting boundary polynomial depends on one variable, or one component of the limiting source gradient takes finitely many values. On the limiting source line, all limiting positions arising from points of E, except for finitely many interval endpoints, lie in the interiors of intervals on which $\psi(t) = \tilde{u}(te)$ is affine. Fenchel's inequality then implies

$$r_{\tau h}(x') \geq c\theta r_h(x)$$

outside finitely many slabs of relative width θ. Covering the slabs by small balls and replacing h by τh on the complement contracts the total weight $r^{n-1+\delta/2}h^{\delta/(2N)}$. The corresponding supremum of normalized Hausdorff content satisfies $M \leq M/2 < \infty$, hence vanishes.

Section 2 records the preliminary facts, and Section 3 establishes the local growth and section estimates. Section 4 proves the selection estimates and fixes the subsets used in the measure comparisons. Section 5 treats the planar limit and the covering argument used in every dimension. Section 6 introduces the graph measure and completes the higher-dimensional proof.

2. Preliminaries

For a proper lower semicontinuous convex function w and $x \in \operatorname{dom} w$, let

$$\partial w(x) = \{p \in \mathbb{R}^n : w(x') \geq w(x) + p \cdot (x' - x)\ \forall x'\} \quad \text{and} \quad w^*(p) = \sup_x\{p \cdot x - w(x)\},$$

and $\operatorname{epi} w = \{(x,t) : t \geq w(x)\}$. Then $w^{**} = w$ and

$$p \in \partial w(x) \iff x \in \partial w^*(p) \iff w(x) + w^*(p) = x \cdot p. \tag{2.1}$$

For a finite convex function w on an open convex set, its Monge-Ampère measure

$$\mu_w(\mathcal{A}) = |\partial w(\mathcal{A})|$$

is a locally finite Borel measure, weakly continuous under local uniform convergence [15, Sections 2.1–2.2]. By Alexandrov's theorem, at almost every x there is a symmetric matrix $D^2w(x) \geq 0$ with

$$w(x + x') = w(x) + Dw(x) \cdot x' + \tfrac{1}{2}x'^{\mathsf{T}} D^2w(x)x' + o(|x'|^2),$$

and the absolutely continuous part of μ_w has density $\det D^2w$. The distributional Hessian of w is a nonnegative matrix-valued Radon measure. Its trace Δw has absolutely continuous part $\operatorname{tr} D^2w$.

Let u be a convex solution to (1.2). We extend u to $\mathbb{R}^n$ as follows

$$\tilde{u}(x) := \sup\{L(x) : L \text{ is affine},\ L \leq u \text{ in } \Omega,\ DL \in \Omega^*\}.$$

The extension $\tilde{u}$ is finite and Lipschitz and agrees with u on Ω. We henceforth denote this extension by u. By mass balance, we have that $Du \in \overline{\Omega^*}$ at every differentiability point in Ω.

The following decomposition follows from Figalli and Kim [16, Proposition 2, the subsequent discussion, and Theorem 3.3] and Zajíček [25, Theorem 1].

Lemma 2.1. *Under the hypotheses of Theorem 1.1,*

$$\Omega' = \{x \in \Omega \setminus A : Du(x) \in \Omega^*\} \quad \textit{and} \quad \Sigma = A \cup E. \tag{2.2}$$

The set A lies in countably many $(n-1)$-dimensional Lipschitz graphs, and $\dim_{\mathrm{H}} A \le n-1$.

Let w be a proper lower semicontinuous convex function, suppose $x_0 \in \partial w(y_0)$, and $h > 0$. The *section* of w at (y_0, x_0) of height h is

$$S_h[w](y_0, x_0) = \{y : w(y) - w(y_0) - x_0 \cdot (y - y_0) < h\}. \tag{2.3}$$

At a differentiability point, write $S_h[w](y_0) = S_h[w](y_0, Dw(y_0))$. The section is *proper* if its closure is compactly contained in the open set on which w is considered.

By John's theorem [17, Theorem III], a bounded convex set $\mathcal{S}$ with nonempty interior has a *normalizing ellipsoid* $z + TB_1$ with

$$z + TB_1 \subset \mathcal{S} \subset z + nTB_1 \quad \text{and} \quad \det T > 0. \tag{2.4}$$

We call T a *normalizing matrix* of $\mathcal{S}$. Up to a rotation of coordinates, T is $\operatorname{diag}(a_1, \dots, a_n)$, with $a_1 \ge \cdots \ge a_n > 0$. Set $\beta = a_1 = \|T\|$, where $\|T\|$ is the operator norm.

The following two estimates will be used later. If w is convex on a bounded convex domain and $\mu_w \ge c_0\, dx$, then every section S_h, not necessarily proper, satisfies [22, Lemma 2.2]

$$|S_h| \le C(n, c_0) h^{n/2}. \tag{2.5}$$

For any proper section of a finite convex function, the Alexandrov estimate [22, Lemma 2.5] implies

$$\mu_w(S_h) \ge c_n h^n / |S_h|. \tag{2.6}$$

3. Localization and section estimates

We first remove one exceptional set and localize the dual potential. On the remaining points, uniform power bounds for u yield uniform estimates for its dual sections. We use only the density bounds in (1.1) until Subsection 3.4.

The argument in Caffarelli [8, Lemma 2], as recalled in [16, Section 3], yields

$$\mu_u \ge (\lambda/\Lambda)\mathbf{1}_\Omega\, dx \qquad \text{in } \mathbb{R}^n. \tag{3.1}$$

3.1. **Pointwise growth.**

Choose a ball B_R containing $\overline{\Omega}$ and consider

$$\widetilde{\mu} = \mu_{u+|\cdot|^2/2}.$$

This measure is finite on B_R and satisfies $\widetilde{\mu} \geq \Delta u$. Indeed, for smooth convex functions,

$$\det(I + D^2 u) \geq \operatorname{tr} D^2 u.$$

Mollification and weak convergence of Monge-Ampère measures [15, Section 2.2] extend this inequality to convex u. For $0 < \alpha < 1$, define

$$Z_\alpha = \left\{ x \in \Omega : \limsup_{r \downarrow 0} \frac{\widetilde{\mu}(B_r(x))}{r^{n-1+\alpha}} = \infty \right\}.$$

For a differentiability point x, write

$$w_x(x') = u(x + x') - u(x) - Du(x) \cdot x'.$$

The measure $\widetilde{\mu}$ controls the lower growth through section volumes and the upper growth through Δu. For the lower bound we use the section argument of Mooney [22, Section 3], keeping track of the distance to one point of the section.

Lemma 3.1. *We have $\mathcal{H}^{n-1+\alpha}(Z_\alpha) = 0$. Every $x \in \Omega \setminus (A \cup Z_\alpha)$ admits $c, C, r_0 > 0$, depending on $x, n, \alpha, \lambda, \Lambda$ such that, for $0 < |x'|, r < r_0$,*

$$c|x'|^N \leq w_x(x') \leq C|x'|^{1+\alpha} \quad \text{and} \quad \partial u(B_r(x)) \subset B_{Cr^\alpha}(Du(x)), \quad \text{where } N = 2/\alpha. \tag{3.2}$$

The supporting plane to u at x touches its graph only at x in Ω.

Proof. The density comparison theorem [23, Chapter 1, Theorem 3.3] gives $\mathcal{H}^{n-1+\alpha}(Z_\alpha) = 0$.

At a point $x \notin Z_\alpha$ there are $M_x \geq 1$ and $0 < r_x < \operatorname{dist}(x, \partial\Omega)$ such that

$$\widetilde{\mu}(B_r(x)) \leq M_x r^{n-1+\alpha} \qquad (0 < r < r_x). \tag{3.3}$$

Fix such a differentiability point and write $w = w_x$, so $w \geq 0$ and $w(0) = 0$.

For $0 < t < r_x$, by a standard smooth approximation and divergence theorem we have

$$m(t) := \frac{1}{|\partial B_t|} \int_{\partial B_t} w \, d\mathcal{H}^{n-1} = \frac{1}{|\mathbb{S}^{n-1}|} \int_0^t \frac{\Delta w(B_s)}{s^{n-1}} \, ds \leq C M_x t^{1+\alpha}.$$

Since $w \geq 0$, by the mean value inequality we have

$$\sup_{B_{2r}} w \leq \frac{1}{|B_r|} \int_{B_{3r}} w \, dz \leq C M_x r^{1+\alpha} \qquad (3r < r_x).$$

This proves the upper growth bound. If $x' \in B_r$ and $0 \neq p \in \partial w(x')$, by the supporting inequality at $x' + rp/|p| \in B_{2r}$ we have

$$r|p| \leq w(x' + rp/|p|) - w(x') \leq \sup_{B_{2r}} w \leq C M_x r^{1+\alpha},$$

from which the subgradient inclusion follows.

Fix a small $x' \neq 0$, choose a sufficiently small height $h > w(x')$. Consider

$$S = \{z : w(z) + \tfrac{1}{2}|z|^2 < h\} \quad \text{and} \quad T = \{z \in B_{r_x} : w(z) < h\}.$$

For small h, $x+S$ is a proper section of $u+|\cdot|^2/2$ and $S\subset B_{\sqrt{2h}}$. The set T is a section of w on the fixed ball B_{r_x}, contains S and x', and satisfies by (3.1) and (2.5),

$$|T|\le Ch^{n/2}. \tag{3.4}$$

Let Q be the projection of S onto $(x')^\perp$. Convexity gives $\frac{1}{2}S+[0,x'/2]\subset T$. It follows that

$$\mathcal{H}^1(T\cap(X+\mathbb{R}x'))\ge\frac{|x'|}{2},\qquad \forall X\in\tfrac{1}{2}Q.$$

On the other hand, $S\subset B_{\sqrt{2h}}$ bounds every line section of S by $2\sqrt{2h}$ in length. By Fubini's theorem,

$$|T|\ge 2^{-n}|x'|\mathcal{H}^{n-1}(Q)\quad\text{and}\quad |S|\le 2\sqrt{2h}\,\mathcal{H}^{n-1}(Q).$$

Combining the above estimates with (3.4) we have

$$|S|\le C\frac{\sqrt{h}}{|x'|}|T|\le C\frac{h^{(n+1)/2}}{|x'|}.$$

It then follows by (2.6) and (3.3) that

$$c|x'|h^{(n-1)/2}\le\frac{c_nh^n}{|S|}\le\widetilde{\mu}(x+S)\le\widetilde{\mu}(B_{\sqrt{2h}}(x))\le CM_xh^{(n-1+\alpha)/2}.$$

Consequently $h\ge(CM_x)^{-2/\alpha}|x'|^{2/\alpha}$. Since this holds for every sufficiently small $h>w(x')$, it follows that

$$w(x')\ge(CM_x)^{-2/\alpha}|x'|^{2/\alpha}.$$

This proves the lower bound and strict convexity at x. □

Splitting the remaining points according to rational bounds for c, C, r_0, we obtain countably many subsets on which (3.2) holds uniformly.

3.2. **A fixed dual potential.**

Using the ball $B_R\supset\overline{\Omega}$ fixed above, define

$$v(y)=\max_{x\in\overline{B_R}}\{x\cdot y-u(x)\}. \tag{3.5}$$

This function is finite, Lipschitz and convex. By biconjugation, $v^*=u$ on B_R [15, Appendix A.4]. We use this same function throughout the proof. The next lemma localizes its equation.

Lemma 3.2. *Let $x_0\in\Omega\setminus(A\cup Z_\alpha)$ and $y_0=Du(x_0)$. There is a bounded neighborhood Ω_0^* of y_0 such that $v=u^*$ on Ω_0^* and $\partial v(\overline{\Omega_0^*})\Subset\Omega$. Moreover $Dv(y_0)=x_0$, and for every Borel set $\mathcal{A}\subset\Omega_0^*$,*

$$\int_{\partial v(\mathcal{A})}f(x)\,dx=\int_{\mathcal{A}\cap\Omega^*}g(y)\,dy. \tag{3.6}$$

In particular,

$$\frac{\lambda}{\Lambda}\mathbf{1}_{\Omega^*}\,dy\le\mu_v\le\frac{\Lambda}{\lambda}\mathbf{1}_{\Omega^*}\,dy\qquad\text{on }\Omega_0^*. \tag{3.7}$$

Proof. Choose $r > 0$ with $\overline{B_{2r}(x_0)} \Subset \Omega$ such that $w_{x_0}(z) \geq c|z|^N$ for $|z| \leq 2r$. Convexity along rays extends this to

$$w_{x_0}(z) \geq cr^{N-1}|z| \qquad (|z| \geq r).$$

Hence, if $|y - y_0| \leq cr^{N-1}/2$, then

$$(y - y_0) \cdot z - w_{x_0}(z) \leq -\tfrac{c}{2} r^{N-1}|z| < 0 \qquad (|z| \geq r),$$

whereas the left side is zero at $z = 0$. Thus, taking $\Omega_0^* = B_{cr^{N-1}/2}(y_0)$, all maximizers lie in $B_r(x_0)$ for every $y \in \overline{\Omega_0^*}$. Consequently, $v = u^*$ on Ω_0^* and $\partial v(\overline{\Omega_0^*}) \subset B_r(x_0) \Subset \Omega$. At y_0 the unique maximizer is x_0, so $Dv(y_0) = x_0$.

Since $v = u^*$ on Ω_0^*, by Legendre duality we have

$$x \in \partial v(\mathcal{A}) \quad \Longleftrightarrow \quad Du(x) \in \mathcal{A}$$

for almost every $x \in \Omega$. Hence by (1.2) and (1.1), we have (3.6) and (3.7). □

Shrink the neighborhoods in Lemma 3.2 to bounded boundary charts Ω_0^* crossing $\partial\Omega^*$. Thus $v = u^*$ on Ω_0^*, $\partial v(\overline{\Omega_0^*}) \Subset \Omega$, and (3.7) holds there. In each chart choose orthonormal coordinates $y = (y', y_n)$ and a smooth function ϱ such that

$$\Omega^* \cap \Omega_0^* = \{y \in \Omega_0^* : y_n > \varrho(y')\}.$$

Define $F(y) = y_n - \varrho(y')$. Then

$$\Omega^* \cap \Omega_0^* = \{F > 0\} \quad \text{and} \quad \partial\Omega^* \cap \Omega_0^* = \{F = 0\}.$$

Since $DF(y) = (-D\varrho(y'), 1)$, we have

$$|DF(y)| = \sqrt{1 + |D\varrho(y')|^2} \geq 1. \tag{3.8}$$

By second countability, countably many such charts cover $Du(E \setminus Z_\alpha)$. On slightly smaller charts the derivatives of ϱ of each fixed order are bounded.

Definition 3.3. Fix $0 < \alpha < 1$, take $N = 2/\alpha$, and choose one such chart Ω_0^*. For fixed $c, C, r_0 > 0$, let $\mathcal{P} = \mathcal{P}(c, C, r_0, \Omega_0^*)$ consist of the points $x \in E \setminus Z_\alpha$ satisfying (3.2) and

$$\overline{B_{r_0}(x)} \subset \Omega \quad \text{and} \quad \overline{B_{r_0}(Du(x))} \subset \Omega_0^*.$$

By Lemmas 3.1 and 3.2, every point of $E \setminus Z_\alpha$ belongs to a set $\mathcal{P}(c, C, r_0, \Omega_0^*)$ from Definition 3.3. Choosing the charts and bounds from countable families therefore gives a countable cover of $E \setminus Z_\alpha$ by such sets. We work on one such set at a time. All subsequent local estimates are uniform in its base point and height.

3.3. **Size of dual sections.**

Fix a set $\mathcal{P}$ as in Definition 3.3 and use v from (3.5). Let T normalize $S_h[v](y_0, x_0)$. Normalizing the target coordinates by T^{-1} and the potential by h^{-1} induces the source coordinate change $X = h^{-1}T^{\mathsf{T}}(x - x_0)$. We call $x_0 + hT^{-\mathsf{T}}B_1$ the *source ellipsoid* associated

with this normalization. Its semiaxes are h/a_i. For $x \in \mathcal{P}$ set

$$\mathcal{R}_h(x) = \sup\{|y - Du(x)| : y \in S_h[v](Du(x), x)\} \quad \text{and} \quad r_h(x) = c_* \frac{h}{\mathcal{R}_h(x)}. \tag{3.9}$$

Here $c_* \in (0, 1]$ depends only on the fixed data and will be chosen in Subsection 5.2. The estimates below hold for any fixed such choice. The radius $r_h(x)$ is comparable to the smallest semiaxis of $x + hT^{-\mathsf{T}}B_1$.

Fix a base point $x_0 \in \mathcal{P}$ and put $y_0 = Du(x_0)$. Translating y_0 and x_0 to the origin and subtracting the supporting affine functions, we may assume $u(0) = v(0) = 0$, $Du(0) = Dv(0) = 0$ and $u, v \geq 0$, so that $S_h = S_h[v](0, 0) = \{v < h\}$. Let $y_h + T_h B_1$ be the normalizing ellipsoid of S_h, with smallest and largest semiaxes $a_{\min}(h)$ and $\beta(h)$, and write $\mathcal{R}_h = \mathcal{R}_h(0)$, $r_h = r_h(0)$.

Lemma 3.4. *Uniformly for base points in $\mathcal{P}$ and for small h, the section S_h is proper and contained in Ω_0^*, and*

$$c_1|y|^{(1+\alpha)/\alpha} \leq v(y) \leq C_1|y|^{N/(N-1)}, \tag{3.10}$$

$$ch^{1-1/N} \leq a_{\min}(h) \leq \beta(h) \leq Ch^{\alpha/(1+\alpha)}, \qquad \beta(h) \leq \mathcal{R}_h \leq 2n\beta(h), \tag{3.11}$$

$$|\det T_h| \geq ch^{n/2}, \tag{3.12}$$

$$ch^{1/(1+\alpha)} \leq r_h \leq Ch^{1/N}, \qquad \|hT_h^{-\mathsf{T}}\| \leq Ch^{1/N}. \tag{3.13}$$

Moreover, $h \mapsto r_h$ is continuous and nondecreasing and attains every sufficiently small positive value.

Proof. By the definition of v and the upper bound in (3.2), we have

$$v(y) \geq \sup_{0<t<r_0} \left(t|y| - Ct^{1+\alpha}\right) = c_1|y|^{(1+\alpha)/\alpha}$$

for small $|y|$. For the upper bound, by the proof of Lemma 3.2, for $|y| < \frac{1}{2}cr_0^{N-1}$ the supremum defining $v(y)$ is therefore attained in B_{r_0}, where it is bounded by $\sup_{t\geq 0}(t|y| - ct^N) = C_1|y|^{N/(N-1)}$. This proves (3.10), from which (3.11) follows.

By (2.6) and (3.7),

$$ch^n/|S_h| \leq \mu_v(S_h) \leq C|S_h|,$$

so $|S_h| \geq ch^{n/2}$. Then, (3.12) follows from $\det T_h \approx |S_h|$. The estimates (3.13) follow from $r_h = c_*h/\mathcal{R}_h$, (3.11) and $\|hT_h^{-\mathsf{T}}\| = h/a_{\min}(h)$.

Since $v(0) = 0$, by convexity we have

$$\frac{h_1}{h_2} S_{h_2} \subset S_{h_1} \subset S_{h_2} \qquad (h_1 < h_2),$$

Hence

$$\frac{h_1}{h_2}\mathcal{R}_{h_2} \leq \mathcal{R}_{h_1} \leq \mathcal{R}_{h_2},$$

so $\mathcal{R}_h$ is continuous and r_h is continuous and nondecreasing. Since $r_h \to 0$, it attains every sufficiently small positive value. □

3.4. **Rescaling the target boundary.**

Recall that F defines the fixed boundary chart. Take $y_0 = Du(x_0)$ with $x_0 \in \mathcal{P}$. Choose an integer d_0 such that

$$d_0 > (1+\alpha)/\alpha. \tag{3.14}$$

Let P_h^0 be the degree-d_0 Taylor polynomial at zero of $Y \mapsto F(y_0 + T_h Y)$, and set

$$\gamma_h = \|P_h^0\|_{L^\infty(B_1)}.$$

Consider a sequence (x_j, y_j, h_j) with $x_j \in \mathcal{P}$, $h_j \downarrow 0$ and $y_j = Du(x_j)$. For each j, let T_j normalize $S_{h_j}[v](y_j, x_j)$, P_j^0 be the degree-d_0 Taylor polynomial at zero of $Y \mapsto F(y_j + T_j Y)$, and $\gamma_j = \|P_j^0\|_{L^\infty(B_1)}$. The following normalization keeps a nonzero polynomial in the limit.

Lemma 3.5. *For every $x_0 \in \mathcal{P}$ and sufficiently small $h > 0$, we have*

$$c a_{\min}(h) \le \gamma_h \le C\beta(h). \tag{3.15}$$

For every sequence as above, there is a subsequence such that, for every $L > 0$,

$$\left\| \frac{F(y_j + T_j \cdot)}{\gamma_j} - P \right\|_{C^2(B_L)} \to 0,$$

where P is a polynomial with $\deg P \le d_0$, $P(0) = 0$ and $\|P\|_{L^\infty(B_1)} = 1$.

Proof. On the space of polynomials of degree at most d_0, the coefficient norm and the $L^\infty(B_1)$ norm are equivalent. Since the linear coefficient of P_h^0 is $T_h^{\mathsf{T}} DF(y_0)$, we find by (3.8) that

$$\gamma_h \ge c|T_h^{\mathsf{T}} DF(y_0)| \ge c a_{\min}(h).$$

Since $F(y_0) = 0$, the constant coefficient vanishes. For $1 \le k \le d_0$, each order-k coefficient is bounded by

$$C\|D^k F(y_0)\| \, \|T_h\|^k \le C\beta(h)^k,$$

since the derivatives of F are uniformly bounded. Thus $\gamma_h \le C\beta(h)$ for small h. All constants are uniform for $x_0 \in \mathcal{P}$ and in the orthonormal frame. Hence (3.15) follows.

Fix a sequence $x_j \in \mathcal{P}$, $h_j \to 0$. Let $\beta_j = \|T_j\|$. By (3.15) and (3.11), $\gamma_j \ge c h_j^{1-1/N}$. For each fixed $L > 0$, this lower bound, Taylor's formula and (3.11) show

$$\left\| \frac{F(y_j + T_j \cdot) - P_j^0}{\gamma_j} \right\|_{C^2(B_L)} \le C_L \frac{\beta_j^{d_0+1}}{\gamma_j} \le C_L h_j^{\alpha(d_0+1)/(1+\alpha) - (1-1/N)} \to 0.$$

Here C_L is independent of j, and the exponent is positive by (3.14). The polynomials P_j^0/γ_j have degree at most d_0, vanish at zero, and have $L^\infty(B_1)$ norm one. Thus $P_j^0/\gamma_j \to P$ in $C^2(B_L)$ for every fixed $L > 0$, with $P(0) = 0$ and $\|P\|_{L^\infty(B_1)} = 1$. □

4. Finite measures and coverings

Fix a set $\mathcal{P} = \mathcal{P}(c, C, r_0, \Omega_0^*)$ as in Definition 3.3. Choose

$$\Omega_0 = \{x \in \Omega : \operatorname{dist}(x, \partial\Omega) > r_0/2\}, \quad \text{and} \quad \Omega_1^* = \{y \in \Omega_0^* : \operatorname{dist}(y, \partial\Omega_0^*) > r_0/2\}.$$

By Definition 3.3,

$$\mathcal{P} \subset \Omega_0 \quad \text{and} \quad Du(\mathcal{P}) \subset \Omega_1^*,$$

and these two regions will contain the source and target ellipsoids, respectively, at sufficiently small heights. For every $x_0 \in \mathcal{P}$, let T_h normalize $S_h[v](y_0, x_0)$ with $y_0 = Du(x_0)$. By Definition 3.3 and Lemma 3.4, there exists $h_0 > 0$, independent of x_0, such that for all

$$0 < h < h_0,$$

there holds

$$S_h[v](y_0, x_0) \Subset \Omega_1^*, \quad x_0 + hT_h^{-\mathsf{T}} B_1 \Subset \Omega_0, \quad \partial u(x_0 + hT_h^{-\mathsf{T}} B_1) \subset \Omega_1^*. \tag{4.1}$$

The last inclusion follows from the uniform subgradient estimate in (3.2), since the source ellipsoid shrinks uniformly and $\operatorname{dist}(Du(x_0), \partial\Omega_0^*) > r_0$.

We henceforth consider only heights $0 < h < h_0$.

4.1. **Weighted measures.**

Let $0 < \epsilon < 1$. Define the target measure ν_0 on Ω_1^* and the source measure μ_0 on Ω_0 by

$$d\nu_0(y) = \mathbf{1}_{\{F(y)\neq 0\}} |F(y)|^{-1+\epsilon}\, dy \quad \text{and} \quad d\mu_0(x) = \mathbf{1}_{\{Du(x)\in\Omega_1^*\cap\Omega^*\}} F(Du(x))^{-1+\epsilon}\, dx. \tag{4.2}$$

The boundary chart and the transport identity imply

$$\nu_0(\Omega_1^*) < \infty \quad \text{and} \quad \mu_0(\Omega_0) \le \frac{\Lambda}{\lambda} \int_{\Omega_1^*\cap\Omega^*} F(y)^{-1+\epsilon}\, dy < \infty. \tag{4.3}$$

For a section of height h with normalizing matrix T and semiaxes $a_1 \ge \cdots \ge a_n$ for the associated John ellipsoid, define the *covering cost* of this section by

$$C_{\mathrm{cov}} := \left(\frac{h}{a_1}\right)^{n-1} \prod_{i>1} \frac{h/a_i}{h/a_1} = \frac{h^{n-1}}{a_2 \cdots a_n}.$$

Here $(h/a_1)^{n-1}$ represents the cost per ball and $\prod_{i>1}(a_1/a_i)$ the number of balls needed to cover the source ellipsoid, both up to dimensional constants.

In the planar proof we use ν_0 and μ_0. In higher dimensions, we retain ν_0 and use the graph measures introduced in Section 6 in place of μ_0.

4.2. **Selection.**

For a section $S_h[v](y_0, x_0)$, let $z + TB_1$ be a normalizing ellipsoid. Write $y = y_0 + TY$ and define

$$\tilde{v}(Y) = \frac{v(y_0 + TY) - v(y_0) - x_0 \cdot TY}{h}. \tag{4.4}$$

We call $\tilde{v}$ the *normalized potential*. Then $\tilde{v} \ge 0 = \tilde{v}(0)$ and $T^{-1}(S_h[v](y_0, x_0) - y_0) = \{\tilde{v} < 1\}$. Set $Y^\circ := T^{-1}(z - y_0)$. By (2.4) and $y_0 \in S_h[v](y_0, x_0)$,

$$B_1(Y^\circ) \subset \{\tilde{v} < 1\} \subset B_{2n} \quad \text{and} \quad |Y^\circ| \le n.$$

We use Macbeath regions [19] and the packing–covering scheme of Bárány and Larman [2]. For epigraph packing in strips of fixed vertical width, see [1]. We give the formulation needed here, including the selected-mass estimate and simultaneous source and target coverings.

Lemma 4.1. *Let v be a proper lower semicontinuous convex function, Ω_1^* a bounded open set, and ν a finite Borel measure on Ω_1^*. Let I be an index set. For $i \in I$, take $x_i \in \partial v(y_i)$, $h_i > 0$, an invertible linear transformation T_i, and a point $Y_i \in \mathbb{R}^n$ in the normalized target coordinates $y = y_i + T_i Y$, subject to the hypotheses below. Define*

$$\tilde{v}_i(Y) = \frac{v(y_i + T_i Y) - v(y_i) - x_i \cdot T_i Y}{h_i}. \tag{4.5}$$

Suppose that, with constants c, C, M, r_1, R_1 uniform in $i \in I$,

$$\sup_{B_{r_1}(Y_i)} \tilde{v}_i \le M \quad \text{and} \quad |Y_i| \le R_1, \tag{4.6}$$

and, for every $i \in I$ and $Z \in \mathbb{R}^n$,

$$\tilde{v}_i(Y_i + Z) + \tilde{v}_i(Y_i - Z) \ge c|Z| - C. \tag{4.7}$$

Then there is $C_* > 0$, depending only on these constants and n, with the following property. For Borel sets $Q_i \subset \Omega_1^* \cap \big(y_i + T_i B_{r_1/20}(Y_i)\big)$ and $h > 0$, take $I_h = \{i \in I : h/2 < h_i \le h\}$. There is a countable $J \subset I_h$ such that

(i) *The selected sets satisfy*

$$\sum_{j \in J} \nu(Q_j) \le C_* \, \nu(\Omega_1^*). \tag{4.8}$$

(ii) *The source and target points are covered by*

$$\{x_i : i \in I_h\} \subset \bigcup_{j \in J} (x_j + C_* h_j T_j^{-\mathsf{T}} B_1) \quad \text{and} \quad \{y_i : i \in I_h\} \subset \bigcup_{j \in J} (y_j + C_* T_j B_1). \tag{4.9}$$

Proof. Step 1. Symmetric regions. Denote $\mathcal{K} = \operatorname{epi} v \subset \mathbb{R}^{n+1}$. For $L_0 \ge M + 2$, let

$$y_i^\circ = y_i + T_i Y_i, \qquad z_i = (y_i^\circ, v(y_i^\circ) + L_0 h_i), \qquad \mathcal{N}_i = (\mathcal{K} - z_i) \cap (z_i - \mathcal{K}).$$

Thus $z_i + \mathcal{N}_i$ is the largest set contained in $\mathcal{K}$ that is centrally symmetric about z_i. Let $\ell_i(y) := v(y_i) + x_i \cdot (y - y_i)$. The coordinates $Y = T_i^{-1}(y - y_i)$ and $s = (t - \ell_i(y))/h_i$ from (4.5) send $\mathcal{K}$ to $\operatorname{epi} \tilde{v}_i$. Let $\widehat{\mathcal{N}}_i$ be the image of $z_i + \mathcal{N}_i$ in these coordinates. By the first estimate in (4.6) and the convexity, we find that

$$B_{r_1}(Y_i) \times [H_i - 1, H_i + 1] \subset \widehat{\mathcal{N}}_i \quad \text{where } H_i = \tilde{v}_i(Y_i) + L_0. \tag{4.10}$$

Take $(Y_i + Z, H_i + \sigma) \in \widehat{\mathcal{N}}_i$, for some $(Z, \sigma) \in \mathbb{R}^n \times \mathbb{R}$. By central symmetry, the reflected point $(Y_i - Z, H_i - \sigma)$ also belongs to $\widehat{\mathcal{N}}_i$. Both points lie in $\operatorname{epi} \tilde{v}_i$, hence

$$\tilde{v}_i(Y_i + Z) \le H_i + \sigma \quad \text{and} \quad \tilde{v}_i(Y_i - Z) \le H_i - \sigma.$$

Since $\tilde{v}_i \geq 0$, these inequalities imply $0 \leq H_i + \sigma \leq 2H_i \leq 2M + 2L_0$, which controls the vertical coordinate of $\widehat{\mathcal{N}_i}$. Adding the above two inequalities and using (4.7) also lead to

$$c|Z| - C \leq \tilde{v}_i(Y_i + Z) + \tilde{v}_i(Y_i - Z) \leq 2H_i \leq 2(M + L_0).$$

Thus $|Z|$ is uniformly bounded. Enlarging C if needed, these bounds give

$$\widehat{\mathcal{N}_i} \subset B_C(Y_i) \times [0, 2M + 2L_0]. \tag{4.11}$$

Taking horizontal projections in (4.10) and (4.11) and returning to the original coordinates,

$$y_i^\circ + T_i B_{r_1} \subset \{y : \text{there exists } t \text{ with } (y,t) \in z_i + \mathcal{N}_i\} \subset y_i^\circ + T_i B_C. \tag{4.12}$$

The vertical interval bounds in (4.10) and (4.11), rescaled by h_i, also give

$$2h_i \leq \mathcal{L}^1\{t : (y,t) \in z_i + \mathcal{N}_i\} \leq Ch_i \qquad (\ \forall y \in y_i^\circ + T_i B_{r_1}\ ). \tag{4.13}$$

Here C depends only on the constants in the hypotheses and the fixed choice of L_0.

Step 2. Selection. In the (Y, s) coordinates, shrinking about z_i acts as $(Z, \sigma) \to (Z/10, \sigma/10)$. Since $Q_i \subset y_i^\circ + T_i B_{r_1/20}$, multiplying its normalized horizontal displacements by 10 places them in $B_{r_1/2}$. Thus (4.10) gives

$$\mathcal{L}^1\{t : (y,t) \in z_i + \tfrac{1}{10} \operatorname{int} \mathcal{N}_i\} \geq h_i/5 \qquad (y \in Q_i). \tag{4.14}$$

For every $(y,t) \in z_i + \frac{1}{10} \operatorname{int} \mathcal{N}_i$, there is a point $(\bar{y}, \bar{t}) \in z_i + \mathcal{N}_i \subset \operatorname{epi} v$ such that

$$(y,t) = \tfrac{9}{10} z_i + \tfrac{1}{10}(\bar{y}, \bar{t}).$$

Using the definition of z_i and convexity, we obtain

$$t - v(y) \geq \big(\frac{9}{10} v(y_i^\circ) + \frac{9}{10} L_0 h_i + \frac{1}{10}\bar{t}\big) - \big(\frac{9}{10} v(y_i^\circ) + \frac{1}{10} v(\bar{y})\big) \geq \frac{9}{10} L_0 h_i.$$

Since both (y,t) and its reflection $\big(2y_i^\circ - y, 2\big(v(y_i^\circ) + L_0 h_i\big) - t\big)$ about z_i lie above ℓ_i, we have

$$t - v(y) \leq t - \ell_i(y) \leq 2\big(v(y_i^\circ) + L_0 h_i - \ell_i(y_i^\circ)\big) \leq 2(M + L_0) h_i. \tag{4.15}$$

Thus, for $h/2 < h_i \leq h$, every $(y,t) \in z_i + \frac{1}{10} \operatorname{int} \mathcal{N}_i$ lies in the same strip

$$ch \leq t - v(y) \leq Ch. \tag{4.16}$$

Choose a maximal pairwise disjoint subfamily of $z_i + \frac{1}{10} \operatorname{int} \mathcal{N}_i$ with $h/2 < h_i \leq h$, indexed by j. It is countable. For each fixed y, their vertical intervals are disjoint and lie in the strip (4.16). Integrating their lengths over $Q_j \subset \Omega_1^*$ and using (4.14), Fubini's theorem and (4.15) in order, we have

$$\begin{aligned} \frac{h}{10} \sum_j \nu(Q_j) &\leq \sum_j \int_{Q_j} \mathcal{L}^1\{t : (y,t) \in z_j + \tfrac{1}{10} \operatorname{int} \mathcal{N}_j\}\, d\nu(y) \\ &\leq \int_{\Omega_1^*} \mathcal{L}^1\{t : ch \leq t - v(y) \leq Ch\}\, d\nu(y) \leq Ch\, \nu(\Omega_1^*). \end{aligned}$$

Dividing by h proves (4.8).

Step 3. Target covering. Fix $i \in I_h$. By maximality in Step 2, there is $j \in J$ such that

$$(z_i + \tfrac{1}{10}\operatorname{int}\mathcal{N}_i) \cap (z_j + \tfrac{1}{10}\operatorname{int}\mathcal{N}_j) \neq \varnothing. \tag{4.17}$$

The corresponding 1/5-regions also intersect. The Macbeath overlap property [5, Lemma 2.5] gives

$$z_i + \tfrac{1}{5}\mathcal{N}_i \subset z_j + \mathcal{N}_j, \qquad z_j + \tfrac{1}{5}\mathcal{N}_j \subset z_i + \mathcal{N}_i. \tag{4.18}$$

To apply the convex-body version, intersect $\mathcal{K}$ with a sufficiently large closed ball containing both full regions, which are bounded by (4.11). This leaves their Macbeath regions unchanged. Taking difference sets in (4.18) and using central symmetry yields

$$\tfrac{1}{5}\mathcal{N}_j \subset \mathcal{N}_i \subset 5\mathcal{N}_j. \tag{4.19}$$

Taking horizontal projections and using (4.12), we obtain

$$cT_jB_1 \subset T_iB_1 \subset CT_jB_1.$$

Also, (4.18) gives $z_i \in z_j + \mathcal{N}_j$, hence $y_i^\circ - y_j^\circ \in CT_jB_1$ by (4.12). Since $y_i^\circ - y_i = T_iY_i$ and $|Y_i|, |Y_j| \le R_1$,

$$y_i - y_j = (y_i^\circ - y_j^\circ) - T_iY_i + T_jY_j \in CT_jB_1.$$

This proves the target covering in (4.9).

Step 4. Source covering. By (4.18),

$$z_j + \tfrac{1}{5}\mathcal{N}_j \subset (z_i + \mathcal{N}_i) \cap (z_j + \mathcal{N}_j). \tag{4.20}$$

By (4.12), its horizontal projection contains

$$y_j^\circ + \tfrac{r_1}{5}T_jB_1. \tag{4.21}$$

On $z_i + \mathcal{N}_i$, reflection about z_i, as in (4.15), gives

$$0 \le v(y) - \ell_i(y) \le t - \ell_i(y) \le 2(M + L_0)h_i.$$

The same bound holds with j in place of i. Since $h_i, h_j \le h$, (4.20) therefore implies

$$|\ell_i(y) - \ell_j(y)| \le Ch \qquad (y \in y_j^\circ + \tfrac{r_1}{5}T_jB_1).$$

The affine function $\ell_i - \ell_j$ has slope $x_i - x_j$, so its oscillation on (4.21) is $\frac{2r_1}{5}|T_j^{\mathsf{T}}(x_i - x_j)|$. As $h < 2h_j$, we obtain

$$|T_j^{\mathsf{T}}(x_i - x_j)| \le Ch_j.$$

Thus $x_i \in x_j + Ch_jT_j^{-\mathsf{T}}B_1$, proving the source covering in (4.9) and completing (ii). □

To control source-side measures, we also apply Lemma 4.1 to v^*. For ordinary sections, write $\tilde{u}_i = \tilde{v}_i^*$ and $x = x_i + h_iT_i^{-\mathsf{T}}X$. Then

$$\tilde{u}_i(X) = \frac{v^*(x_i + h_iT_i^{-\mathsf{T}}X) - v^*(x_i) - y_i \cdot h_iT_i^{-\mathsf{T}}X}{h_i} \ge 0, \tag{4.22}$$
$$X \in \partial\tilde{v}_i(Y) \iff Y \in \partial\tilde{u}_i(X).$$

Let $b_i + T_i B_1$ be the chosen normalizing ellipsoid and set $Y_i := T_i^{-1}(b_i - y_i)$. By (2.4),

$$B_1(Y_i) \subset \{\tilde{v}_i < 1\} \subset B_{2n} \quad \text{and} \quad |Y_i| \le n.$$

Convexity along $\{\lambda Y : \lambda \ge 0\}$ gives

$$\tilde{v}_i(Y) \ge |Y|/(2n) - 1. \tag{4.23}$$

Since $\tilde{u}_i = \tilde{v}_i^*$, we thus see from the definition of Legendre transform that

$$0 \le \tilde{u}_i(X) \le \sup_{Y \in \mathbb{R}^n} \left\{ X \cdot Y - \frac{|Y|}{2n} + 1 \right\} = 1 \qquad \text{for } X \in B_{1/(2n)}. \tag{4.24}$$

Since $B_1(Y_i) \subset \{\widetilde{v}_i < 1\}$, we have

$$\tilde{u}_i(X) \ge \sup_{Y \in B_1(Y_i)} X \cdot Y - 1 = Y_i \cdot X + |X| - 1.$$

Switching X and $-X$ then shows

$$\tilde{u}_i(X) + \tilde{u}_i(-X) \ge 2|X| - 2. \tag{4.25}$$

We now fix the constant ρ as below which is used throughout the paper,

$$0 < \rho < \frac{1}{40n}. \tag{4.26}$$

Together with (4.24), (4.25) verifies the hypotheses of Lemma 4.1 for v^* with base points x_i, supporting slopes y_i, normalizing matrices $h_i T_i^{-\mathsf{T}}$ and normalized centers zero. The local boundedness radius is $1/(2n)$, so (4.26) verifies the prescribed set condition. For a finite source measure μ and prescribed Borel subsets of $x_i + \rho h_i T_i^{-\mathsf{T}} B_1$, the lemma selects a countable subfamily with bounded total μ-mass and covers the source centers by $x_j + C h_j T_j^{-\mathsf{T}} B_1$.

For $s > 0$, let c_s be the standard normalization constant for $\mathcal{H}^s$. For $\mathcal{A} \subset \mathbb{R}^n$ and $0 < \eta \le \infty$, define

$$\mathcal{H}^s_\eta(\mathcal{A}) = c_s \inf \left\{ \sum_{i=1}^{\infty} (\operatorname{diam} U_i)^s : \mathcal{A} \subset \bigcup_{i=1}^{\infty} U_i,\ \operatorname{diam} U_i \le \eta \right\}.$$

The infimum is over countable covers by sets $U_i \subset \mathbb{R}^n$. The case $\eta = \infty$ imposes no diameter restriction and gives the s-dimensional Hausdorff content $\mathcal{H}^s_\infty$. The Hausdorff measure satisfies $\mathcal{H}^s(\mathcal{A}) = \lim_{\eta \downarrow 0} \mathcal{H}^s_\eta(\mathcal{A})$, with the normalization chosen so that $\mathcal{H}^k$ agrees with k-dimensional Lebesgue measure on every k-dimensional affine subspace.

The next lemma bounds the Hausdorff covering cost of a source ellipsoid by the measure of a corresponding set.

Lemma 4.2. *Let $\mathcal{E}$ be an ellipsoid with semiaxes $b_1 \ge \cdots \ge b_n$, and let $\delta \ge 0$. Take*

$$C_{\mathrm{cov}} = \prod_{i<n} b_i.$$

(i) *The geometric covering estimate is*

$$\mathcal{H}^{n-1+\delta}_{2b_n}(\mathcal{E}) \le C C_{\mathrm{cov}}\, b_n^{\delta}.$$

(ii) *If $\mathcal{E}$ is the source ellipsoid associated with a section of v at height h, with normalizing matrix T and $\beta = \|T\|$, then $b_n = h/\beta \le C\sqrt{h}$. If a measure μ and a set Q satisfy $\mu(Q) \ge ch^\epsilon C_{\mathrm{cov}}$, then*

$$\mathcal{H}^{n-1+\delta}_{C\sqrt{h}}(\mathcal{E}) \le Ch^{\delta/2-\epsilon}\mu(Q). \tag{4.27}$$

Proof. A grid of mesh $b_n/\sqrt{n}$ in the principal frame covers $\mathcal{E}$ by at most $C\prod_{i<n}(b_i/b_n)$ balls of radius b_n. Thus

$$\mathcal{H}^{n-1+\delta}_{2b_n}(\mathcal{E}) \le C\prod_{i<n}(b_i/b_n)b_n^{n-1+\delta} = CC_{\mathrm{cov}}b_n^\delta.$$

For a section of v, (3.12) gives

$$\beta^n \ge |\det T| \ge ch^{n/2},$$

so $b_n = h/\beta \le C\sqrt{h}$. If $\mu(Q) \ge ch^\epsilon C_{\mathrm{cov}}$, then

$$\mathcal{H}^{n-1+\delta}_{C\sqrt{h}}(\mathcal{E}) \le CC_{\mathrm{cov}}\, h^{\delta/2} \le Ch^{\delta/2-\epsilon}\mu(Q).$$

□

4.3. **The normalized mass.**

Let $x \in \mathcal{P}$, $y = Du(x)$, and let $b_h + T_hB_1$ be a normalizing ellipsoid of $S_h[v](y,x)$. Let $\tilde{v}$ be the normalized potential in (4.4) for this section, and let $\tilde{u} = \tilde{v}^*$. John's normalization (2.4), (4.23) and (4.25) verify the selection hypotheses for $\tilde{v}$ and $\tilde{u}$. With ρ fixed by (4.26), set

$$\Omega^*_h(x) = b_h + \rho T_hB_1, \qquad \Omega_h(x) = x + \rho h T_h^{-\mathsf{T}}B_1, \qquad C_{\mathrm{cov}}(x,h) = \frac{\|T_h\|h^{n-1}}{|\det T_h|}. \tag{4.28}$$

By (4.26), Lemma 4.1 and its source-side version apply to the prescribed sets $\Omega^*_h(x)$ and $\Omega_h(x)$, respectively. Let ν be a finite Borel measure on Ω^*_1 and μ a finite Borel measure on Ω_0. Define the *normalized mass* at x of the scale h by

$$\Theta_h(x) = \frac{\nu(\Omega^*_h(x)) + \mu(\Omega_h(x))}{h^\epsilon C_{\mathrm{cov}}(x,h)}. \tag{4.29}$$

Throughout Section 5 we will take $\nu = \nu_0$ and $\mu = \mu_0$. In Section 6 we keep ν_0, replace μ_0 by a graph projection, and add one term from the graph itself.

Consider the following partition of $\mathcal{P}$:

$$\mathcal{P}_+ = \{x \in \mathcal{P} : \limsup_{h\downarrow 0}\Theta_h(x) > 0\} \quad \text{and} \quad \mathcal{P}_0 = \{x \in \mathcal{P} : \lim_{h\downarrow 0}\Theta_h(x) = 0\}. \tag{4.30}$$

Proposition 4.3. *For the finite measures in* (4.29), $\dim_{\mathrm{H}}\mathcal{P}_+ \le n - 1 + 2\epsilon$.

Proof. Fix $\delta > 2\epsilon$. For every $x \in \mathcal{P}_+$, there is a positive rational b such that $\Theta_h(x) > 2b$ at arbitrarily small heights. At each such height, at least one of the following inequalities holds:

$$\nu(\Omega^*_h(x)) \ge bh^\epsilon C_{\mathrm{cov}}(x,h) \quad \text{or} \quad \mu(\Omega_h(x)) \ge bh^\epsilon C_{\mathrm{cov}}(x,h). \tag{4.31}$$

Along a sequence of these heights tending to zero, one inequality holds infinitely often. It therefore suffices, for each fixed b, to treat separately the points admitting arbitrarily small heights with the first or the second inequality in (4.31).

Consider the first inequality in (4.31). For each k, we define

$$I_{2^{-k}} = \{(x,t) : x \in \mathcal{P},\ 2^{-k-1} < t \le 2^{-k},\ \nu(\Omega_t^*(x)) \ge bt^\epsilon C_{\mathrm{cov}}(x,t)\}.$$

This is the height group I_h in Lemma 4.1 with $h = 2^{-k}$. For $i = (x,t) \in I_{2^{-k}}$, set $x_i = x$, $h_i = t$ and $y_i = Du(x_i)$, and let $b_i + T_iB_1$ normalize $S_{h_i}[v](y_i, x_i)$. Apply Lemma 4.1 to v with these data, centers $Y_i = T_i^{-1}(b_i - y_i)$, and prescribed sets $Q_i = \Omega^*_{h_i}(x_i) = b_i + \rho T_iB_1$ from (4.28). The hypotheses (4.6) and (4.7) follow from (2.4) and (4.23). Since $\rho < 1/20$, for all sufficiently large k, the section containment in (4.1) gives

$$Q_i = y_i + T_iB_\rho(Y_i) \subset \Omega_1^* \cap (y_i + T_iB_{1/20}(Y_i)).$$

The lemma therefore selects a countable $J \subset I_{2^{-k}}$. The subfamily $(Q_j)_{j\in J}$ has total ν-mass bounded by (4.8), and (4.9) covers all qualifying source points. Denote these ellipsoids by $\mathcal{E}_j = x_j + C_*h_jT_j^{-\mathsf{T}}B_1$. By (4.27),

$$\sum_j \mathcal{H}^{n-1+\delta}_{C2^{-k/2}}(\mathcal{E}_j) \le C_b \sum_j h_j^{\delta/2-\epsilon}\nu(Q_j) \le C_b 2^{-k(\delta/2-\epsilon)} \sum_j \nu(Q_j) \le C_b' 2^{-k(\delta/2-\epsilon)}. \tag{4.32}$$

Here $\delta/2 - \epsilon > 0$. Let $\mathcal{P}_{+,b}$ be the set of points $x \in \mathcal{P}$ such that the first inequality in (4.31) holds infinitely often along a sequence of heights tending to zero. For each sufficiently large k, denote the source ellipsoids selected above by $\mathcal{E}_{k,j}$. Then for every sufficiently large integer k_0 there holds

$$\mathcal{P}_{+,b} \subset \bigcup_{k\ge k_0} \bigcup_j \mathcal{E}_{k,j}.$$

It follows that

$$\mathcal{H}^{n-1+\delta}_{C2^{-k_0/2}}(\mathcal{P}_{+,b}) \le \sum_{k\ge k_0} \sum_j \mathcal{H}^{n-1+\delta}_{C2^{-k/2}}(\mathcal{E}_{k,j}) \le C_b' \sum_{k\ge k_0} 2^{-k(\delta/2-\epsilon)},$$

where (4.32) is used in the last inequality. Since k_0 can be arbitrarily large, $\mathcal{H}^{n-1+\delta}(\mathcal{P}_{+,b}) = 0$.

For the second inequality in (4.31), apply the lemma to v^* with the dual data in (4.22), measure μ, and prescribed sets $Q_i = \Omega_{h_i}(x_i)$ from (4.28). Estimate (4.25) verifies the symmetric growth hypothesis. The local boundedness radius $1/(2n)$ in (4.24) and (4.26) verify the prescribed-set condition. The same selection and covering argument applies. Taking the union of the two cases over positive rational b gives $\mathcal{H}^{n-1+\delta}(\mathcal{P}_+) = 0$. Letting $\delta \downarrow 2\epsilon$ proves the conclusion. □

5. The two-dimensional case

Throughout this section $n = 2$, and Θ_h uses the weighted measures in (4.2), which are finite by (4.3). We estimate the product of their masses on $\Omega_h^*(x)$ and $\Omega_h(x)$.

5.1. **Limits of normalized sections.**

Let $x_j \in \mathcal{P}$, $y_j = Du(x_j)$ and $h_j \downarrow 0$. Write

$$\tilde v_j(Y) = \frac{v(y_j + T_jY) - v(y_j) - x_j \cdot T_jY}{h_j} \quad \text{and} \quad \tilde u_j = \tilde v_j^*, \tag{5.1}$$

where T_j normalizes $S_{h_j}[v](y_j, x_j)$. Let $b_j + T_j B_1$ be the corresponding normalizing ellipsoid and set $Y_j^0 := T_j^{-1}(b_j - y_j)$. Then $|Y_j^0| \le n$ and

$$\tilde{v}_j \ge 0, \qquad \tilde{v}_j(0) = 0, \qquad B_1(Y_j^0) \subset \{\tilde{v}_j < 1\} \subset B_{2n}. \tag{5.2}$$

Choose orthonormal bases in the domain and range of T_j in which T_j is diagonal, and pass to a subsequence on which these bases converge. Put

$$\beta_j = a_{1,j} \ge a_{2,j} > 0, \qquad C_{\mathrm{cov},j} = \frac{\beta_j h_j}{|\det T_j|}, \qquad F_j(Y) = F(y_j + T_j Y).$$

By Lemma 3.5, after a subsequence there are positive numbers γ_j and a nonzero polynomial P of degree at most d_0 such that

$$P_j := F_j/\gamma_j \to P \quad \text{in } C^2 \text{ on bounded sets} \quad \text{and} \quad P(0) = 0. \tag{5.3}$$

Proposition 5.1. *Let $n = 2$ and suppose $\Theta_{h_j}(x_j) \to 0$. After passing to a subsequence,*

$$T_j/\beta_j \to e^* \otimes e \quad \textit{and} \quad P(Y) = p(e \cdot Y),$$

where $|e^*| = |e| = 1$ and p is a nonzero polynomial of degree at most d_0. For the fixed small ρ chosen in (4.26)*, there is a convex function $\tilde{u}$ on $B_{2\rho}$ such that $\tilde{u}_j \to \tilde{u}$ locally uniformly there. The subgradients of $\tilde{u}_j$ are uniformly bounded on the smaller ball B_ρ.*

Proof. Step 1. Compactness. By (4.24), $0 \le \tilde{u}_j \le 1$ on $B_{1/(2n)}$. By (4.26), $4\rho < 1/(2n)$. The convex subgradient estimate and Arzelà-Ascoli yield a subsequence with $\tilde{u}_j \to \tilde{u}$ locally uniformly on $B_{2\rho}$ and uniformly bounded subgradients on this ball.

Step 2. Rank one limit. By (5.3), $|F_j| \le C\gamma_j$ on each fixed ball. Since the normalized target ellipsoids have bounded centers and fixed radius ρ, and $\{F = 0\}$ has Lebesgue measure zero,

$$\nu_0(\Omega^*_{h_j}(x_j)) \ge c\gamma_j^{-1+\epsilon} |\det T_j|. \tag{5.4}$$

For almost every $X \in B_\rho$, put $x = x_j + h_j T_j^{-\mathsf{T}} X$. The source normalization (4.22) gives

$$Du(x) = y_j + T_j D\tilde{u}_j(X). \tag{5.5}$$

By the definition of F_j, we have $F(Du(x)) = F_j(D\tilde{u}_j(X))$. So (5.3) shows $|F(Du(x))| \le C\gamma_j$. Changing variables from x to X, we obtain

$$\mu_0(\Omega_{h_j}(x_j)) \ge c\gamma_j^{-1+\epsilon} \frac{h_j^2}{|\det T_j|}. \tag{5.6}$$

Since $C_{\mathrm{cov},j} = h_j/a_{2,j}$, multiplying (5.4) and (5.6) gives

$$c\left(\frac{\gamma_j}{h_j}\right)^{2\epsilon} \frac{a_{2,j}^2}{\gamma_j^2} \le \frac{\nu_0(\Omega^*_{h_j}(x_j))\mu_0(\Omega_{h_j}(x_j))}{h_j^{2\epsilon} C_{\mathrm{cov},j}^2} \le \tfrac{1}{4}\Theta_{h_j}(x_j)^2 \to 0. \tag{5.7}$$

By (3.15) and (3.11), $\gamma_j \ge ch_j$ for $h_j \le 1$. Together with $\gamma_j \le C\beta_j$, this proves

$$\frac{a_{2,j}}{\gamma_j} \to 0 \quad \text{and} \quad \frac{a_{2,j}}{\beta_j} \to 0.$$

Choose orthonormal bases $e_{1,j}, e_{2,j}$ and $e^*_{1,j}, e^*_{2,j}$ such that $T_j e_{i,j} = a_{i,j} e^*_{i,j}$ for $i = 1, 2$. After passing to a further subsequence, both bases converge. Define

$$e = \lim_j e_{1,j}, \qquad e_2 = \lim_j e_{2,j}, \qquad e^* = \lim_j e^*_{1,j}.$$

Then e, e_2 are orthonormal, $|e^*| = 1$, and

$$\frac{T_j}{\beta_j} = e^*_{1,j} \otimes e_{1,j} + \frac{a_{2,j}}{\beta_j} e^*_{2,j} \otimes e_{2,j} \to e^* \otimes e.$$

Step 3. Boundary polynomial. On each fixed ball, the chain rule gives

$$|DP_j(Y) \cdot e_{2,j}| = \frac{|DF(y_j + T_j Y) \cdot T_j e_{2,j}|}{\gamma_j} \le C \frac{a_{2,j}}{\gamma_j} \to 0.$$

Passing to the limit yields $DP \cdot e_2 = 0$. Thus P is constant along lines parallel to e_2, and

$$P(Y) = p(e \cdot Y) \quad \text{and} \quad p(t) := P(te) \quad (t \in \mathbb{R}).$$

Since P is a nonzero polynomial of degree at most d_0, so is p. □

5.2. **Comparison of radii.**

We next compare the radii of sections centered at nearby points. The argument works in every dimension and will also be used in Section 6.

Lemma 5.2. *Let $\tilde{v} \ge 0$ be convex, $\tilde{v}(0) = 0$, and suppose $B_{r_1}(Y^\circ) \subset \{\tilde{v} < 1\} \subset B_{R_1}$ for fixed $r_1, R_1 > 0$. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear map and set $\beta = \|T\|$. There are $c, C > 0$, depending only on r_1, R_1, such that*

$$c\beta \le \sup_{Y \in \overline{S_1[\tilde{v}](Y', X)}} |T(Y - Y')| \le C\beta, \quad \forall X \in B_{1/(4R_1)}, \ \forall Y' \in \partial \tilde{v}^*(X).$$

Proof. Take $X \in B_{1/(4R_1)}$ and $Y' \in \partial \tilde{v}^*(X)$. By duality, $X \in \partial \tilde{v}(Y')$ and $\tilde{v}(Y') \le X \cdot Y'$. It follows from convexity that

$$\tilde{v}(Y) \ge |Y|/R_1 \qquad \text{for } |Y| \ge R_1. \tag{5.8}$$

Therefore $|Y'| < R_1$. Otherwise the preceding estimates imply $|X| \ge R_1^{-1}$, a contradiction.

For $Y \in \overline{S_1[\tilde{v}](Y', X)}$ with $|Y| \ge R_1$, we infer from (5.8) and $|X| \le (4R_1)^{-1}$ that

$$1 \ge \tilde{v}(Y) - \tilde{v}(Y') - X \cdot (Y - Y') \ge \tilde{v}(Y) - X \cdot Y \ge \frac{3|Y|}{4R_1}.$$

Hence $S_1[\tilde{v}](Y', X) \subset B_{4R_1/3}$. As a result, for every $Y \in S_1[\tilde{v}](Y', X)$,

$$|T(Y - Y')| \le \beta(|Y| + |Y'|) \le C_{R_1}\beta.$$

Let e be a unit vector such that $|Te| = \beta$. Then $|TZ| \ge \beta |e \cdot Z|$ for every $Z \in \mathbb{R}^n$. Choose $Y_1 \in B_{r_1}(Y^\circ)$ such that $|e \cdot (Y_1 - Y')| \ge r_1/2$. Since $Y_1, Y' \in B_{R_1}$,

$$\tilde{v}(Y_1) - \tilde{v}(Y') - X \cdot (Y_1 - Y') \le 1 + |X||Y_1 - Y'| \le 1 + \frac{2R_1}{4R_1} = \frac{3}{2}.$$

By convexity, $(Y'+Y_1)/2 \in S_1[\tilde{v}](Y',X)$. The lower bound follows, since

$$\left|T\left(\frac{Y'+Y_1}{2}-Y'\right)\right| \geq \frac{\beta}{2}|e\cdot(Y_1-Y')| \geq \frac{r_1\beta}{4}.$$

□

With ρ fixed by (4.26), we now choose $0 < c_* \leq \rho/4$ in (3.9) and fix it throughout the paper. If T normalizes the section at $(Du(x),x)$, for every $x' \in B_{r_h(x)}(x)$, write

$$x' = x + \frac{h}{\beta}\widehat{X} \quad \text{and} \quad X = \frac{T^{\mathsf{T}}\widehat{X}}{\beta},$$

By (3.9) and the inequality $\beta \leq \mathcal{R}_h(x)$ proved in Lemma 3.4,

$$|X| \leq |\widehat{X}| \leq \frac{\beta r_h(x)}{h} = c_*\frac{\beta}{\mathcal{R}_h(x)} \leq c_* \leq \rho/4. \tag{5.9}$$

Thus X lies in a fixed ball compactly contained in B_ρ.

Let $\tilde{v}$ be the potential normalized as in (4.4) at $(Du(x),x)$. For $x' \in \mathcal{P} \cap B_{r_h(x)}(x)$, let

$$Y' := T^{-1}(Du(x') - Du(x)).$$

By (4.22), $Y' \in \partial\tilde{v}^*(X)$. Since $y = Du(x) + TY$, the recentered sections satisfy

$$S_h[v](Du(x'),x') = Du(x) + TS_1[\tilde{v}](Y',X).$$

Hence Lemma 5.2, applied to $\tilde{v}$ with inner radius 1 and outer radius $2n$, shows

$$c\beta \leq \mathcal{R}_h(x') = \sup_{Y\in\overline{S_1[\tilde{v}](Y',X)}} |T(Y-Y')| \leq C\beta.$$

Since $\beta \leq \mathcal{R}_h(x) \leq 2n\beta$ by (3.11) and $r_h = c_*h/\mathcal{R}_h$ by (3.9), we obtain

$$c_2 r_h(x) \leq r_h(x') \leq C_2 r_h(x) \qquad (x' \in \mathcal{P} \cap B_{r_h(x)}(x)). \tag{5.10}$$

By Lemma 3.4, $h \mapsto r_h(x)$ is continuous and nondecreasing, and

$$r_h(x) \leq C_0 h^{1/N}. \tag{5.11}$$

In particular, these radii tend to zero uniformly on $\mathcal{P}$.

The next proposition converts the sequential limit classification into uniform small-scale radius estimates, without choosing a limit model for each base point and height.

Proposition 5.3. *Let $n \geq 2$. Assume that whenever $x_j \in \mathcal{P}$, $h_j \downarrow 0$ and $\Theta_{h_j}(x_j) \to 0$, there is a subsequence for which*

$$\tilde{u}_j \to \tilde{u} \quad \text{locally uniformly on } B_{2\rho}, \qquad T_j/\beta_j \to e^* \otimes e, \qquad |e^*| = |e| = 1, \tag{5.12}$$

and one of the following alternatives holds:

1. *The normalized boundary functions F_j/γ_j converge locally uniformly to $P(Y) = p(e\cdot Y)$, where p is nonzero and has degree $\leq d_0$.*
2. *$\#\{D_e\tilde{u}(X) : X \in B_\rho,\ \tilde{u}$ is differentiable at $X\} \leq d_0$.*

Then for each sufficiently small $\theta > 0$ and each $\tau \in (0,1)$, there are positive thresholds $h_* = h_*(\theta,\tau)$ and $\Theta_* = \Theta_*(\theta,\tau)$ such that, whenever $x \in \mathcal{P}$, $h < h_*$ and $\Theta_h(x) < \Theta_*$, there is an exceptional set $\mathcal{E}_{x,h} \subset \mathcal{P} \cap B_{r_h(x)}(x)$, satisfying the following properties:

(i) *Every subset $\mathcal{A} \subset \mathcal{E}_{x,h}$ can be covered by at most $C\theta^{-(n-1)}$ balls of radius $\theta r_h(x)$ with centers in $\mathcal{A}$.*

(ii) *For every $x' \in (\mathcal{P} \cap B_{r_h(x)}(x)) \setminus \mathcal{E}_{x,h}$,*

$$c\theta r_h(x) \le r_{\tau h}(x') \le C r_h(x). \tag{5.13}$$

The choice of $\mathcal{E}_{x,h}$ also depends on the fixed θ, τ. The constants c, C depend only on n, d_0 and the fixed ρ, c_*, but are independent of θ, τ, x, h and $\mathcal{A}$.

Proof. Fix θ, τ. Assume no thresholds h_*, Θ_* exist. Choose $x_j \in \mathcal{P}$ and $h_j \downarrow 0$, with $\Theta_{h_j}(x_j) < 1/j$, such that no exceptional set satisfies both (i) and (ii). By the assumed sequential property, pass to a subsequence satisfying (5.12) and alternative (1) or (2). We construct exceptional sets $\mathcal{E}_{x_j,h_j}$ satisfying both conclusions for all large j, obtaining a contradiction.

Step 1. Exceptional hyperplanes. In case (1), let $\mathcal{T}$ be the set of real roots of p. In case (2),

$$\mathcal{T} := \{e \cdot D\tilde{u}(X) : X \in B_\rho,\ \tilde{u} \text{ is differentiable at } X\}.$$

In either case, $\#\mathcal{T} \le d_0$. Set $\mathcal{I} = (-\rho/2, \rho/2)$ and $\psi(t) = \tilde{u}(te)$ for $t \in \mathcal{I}$.

Take any sequence $x_j' \in \mathcal{P} \cap B_{r_{h_j}(x_j)}(x_j)$ and define

$$\widehat{X}_j := \frac{\beta_j}{h_j}(x_j' - x_j) \quad \text{and} \quad X_j := \frac{T_j^{\mathsf{T}} \widehat{X}_j}{\beta_j}.$$

By (5.9), $|X_j| \le |\widehat{X}_j| \le \rho/4$. After passing to a subsequence, $\widehat{X}_j \to \widehat{X}$, and (5.12) implies

$$X_j \to (e^* \cdot \widehat{X})e.$$

Since $|\widehat{X}| \le \rho/4$, we have $e^* \cdot \widehat{X} \in [-\rho/4, \rho/4] \Subset \mathcal{I}$. We claim that

$$\partial\psi(e^* \cdot \widehat{X}) \cap \mathcal{T} \neq \varnothing. \tag{5.14}$$

In case (1), set $Y_j := T_j^{-1}(Du(x_j') - y_j)$. By (4.22), $Y_j \in \partial\tilde{u}_j(X_j)$. Local uniform convergence controls these subgradients, so a subsequence satisfies

$$Y_j \to Y \in \partial\tilde{u}((e^* \cdot \widehat{X})e).$$

Since $\mathcal{P} \subset E$, these target points satisfy $F_j(Y_j) = 0$. By (5.3), $p(e \cdot Y) = 0$. Restricting the supporting inequality for $\tilde{u}$ to the line te gives $e \cdot Y \in \partial\psi(e^* \cdot \widehat{X})$, proving (5.14).

In case (2), for each $t \in \mathcal{I}$, approach te by differentiability points of $\tilde{u}$ in B_ρ. A subsequence of the gradients converges to $q \in \partial\tilde{u}(te)$, with $e \cdot q \in \mathcal{T}$. Restricting the supporting inequality to the line $\mathbb{R}e$ gives $\partial\psi(t) \cap \mathcal{T} \neq \varnothing$. Taking $t = e^* \cdot \widehat{X}$ proves (5.14).

For $s \in \mathcal{T}$, consider

$$\mathcal{I}_s = \{t \in \mathcal{I} : s \in \partial\psi(t)\}.$$

Each nonempty $\mathcal{I}_s$ is a relatively closed interval on which ψ is affine with slope s. Denote the endpoints of these intervals lying in $\mathcal{I}$ by t_i. The intervals $\mathcal{I}_s$ need not meet. Since $\mathcal{T}$ contains at most d_0 elements, there are at most $2d_0$ such endpoints. By (5.14), every limiting position lies over one of the intervals $\mathcal{I}_s$. Thus, away from the hyperplanes $\{\widehat{X} : e^* \cdot \widehat{X} = t_i\}$, it lies in the interior of an interval on which ψ is affine.

Step 2. Exceptional covering. Take ϑ to be a sufficiently small fixed multiple of θ. Define

$$\mathcal{E}_{x_j,h_j} = \Big\{x' \in \mathcal{P} \cap B_{r_{h_j}(x_j)}(x_j) : \Big|\frac{\beta_j}{h_j} e^* \cdot (x' - x_j) - t_i\Big| \le 2\vartheta \text{ for some } i\Big\}.$$

By (3.9) and (3.11), h_j/β_j is comparable to $r_{h_j}(x_j)$. Thus $\mathcal{E}_{x_j,h_j}$ lies in at most $2d_0$ slabs of width $C\vartheta r_{h_j}(x_j)$. For each slab, its intersection with $B_{r_{h_j}(x_j)}(x_j)$ can be covered by at most $C\vartheta^{-(n-1)}$ balls of radius $C\vartheta r_{h_j}(x_j)$. Choose the fixed multiple so that this radius is at most $\theta r_{h_j}(x_j)/2$. Given $\mathcal{A} \subset \mathcal{E}_{x_j,h_j}$, retain only those balls in the preceding cover that meet $\mathcal{A}$. For each retained ball, choose a point of $\mathcal{A}$ inside it as the new center. Doubling the radii gives (i).

Step 3. Radius estimates. The upper bound follows from $0 < \tau < 1$ and (5.10):

$$r_{\tau h_j}(x') \le r_{h_j}(x') \le C_2 r_{h_j}(x_j).$$

For the lower bound, take a sequence of points outside $\mathcal{E}_{x_j,h_j}$. Pass to a subsequence with normalized positions $\widehat{X}_j \to \widehat{X}$, and set $t_0 = e^* \cdot \widehat{X}$. Step 1 and the definition of $\mathcal{E}_{x_j,h_j}$ imply that ψ is affine on $[t_0 - \vartheta, t_0 + \vartheta]$. The corresponding target points satisfy, after a subsequence,

$$X_j = \frac{T_j^{\mathsf{T}} \widehat{X}_j}{\beta_j} \to t_0 e, \qquad Y_j \in \partial\tilde{u}_j(X_j), \qquad Y_j \to Y_0 \in \partial\tilde{u}(t_0 e).$$

Since ψ is affine on this interval, $e \cdot Y_0 = \psi'(t_0)$. Hence the nonnegative quantities

$$d_j^{\pm} = \tilde{u}_j(X_j \pm \vartheta e) - \tilde{u}_j(X_j) \mp \vartheta e \cdot Y_j \to 0.$$

The normalized target section at x' is $S_\tau[\tilde{v}_j](Y_j, X_j)$. Since $\tilde{u}_j = \tilde{v}_j^*$ and $Y_j \in \partial\tilde{u}_j(X_j)$,

$$\tilde{v}_j(Y) \ge (X_j \pm \vartheta e) \cdot Y - \tilde{u}_j(X_j \pm \vartheta e) \quad \text{and} \quad \tilde{v}_j(Y_j) = X_j \cdot Y_j - \tilde{u}_j(X_j).$$

Subtracting the equality from the inequality and rearranging with the definition of $d_j^{\pm}$ gives

$$\tilde{v}_j(Y) - \tilde{v}_j(Y_j) - X_j \cdot (Y - Y_j) \ge \pm\vartheta e \cdot (Y - Y_j) - d_j^{\pm}.$$

Hence

$$Y \in S_\tau[\tilde{v}_j](Y_j, X_j) \quad \Longrightarrow \quad |e \cdot (Y - Y_j)| \le \frac{\tau + \max(d_j^+, d_j^-)}{\vartheta}. \tag{5.15}$$

The proof of Lemma 5.2 bounds these sections in a fixed ball. Using (5.15) and (5.12),

$$\frac{\mathcal{R}_{\tau h_j}(x')}{\beta_j} = \sup_{Y \in S_\tau[\tilde{v}_j](Y_j, X_j)} \left|\frac{T_j}{\beta_j}(Y - Y_j)\right| \le \frac{\tau}{\vartheta} + o(1) \le \frac{2\tau}{\vartheta}$$

for large j, with ϑ, τ fixed. By (3.9) and (3.11), $r_{h_j}(x_j) \le c_* h_j/\beta_j$. Hence

$$r_{\tau h_j}(x') = \frac{c_* \tau h_j}{\mathcal{R}_{\tau h_j}(x')} \ge \frac{c_* \vartheta h_j}{2\beta_j} \ge \frac{\vartheta}{2} r_{h_j}(x_j). \tag{5.16}$$

This estimate holds uniformly outside $\mathcal{E}_{x_j,h_j}$ for large j. Otherwise, a sequence of violating points would have a convergent subsequence to which the preceding argument applies. Since ϑ is a fixed multiple of θ, this proves (ii). The constants in (i) and (ii) are independent of θ, τ. Both conclusions now hold for large j, contradicting the choice of the sequence. □

5.3. **The covering argument.**

Proposition 5.4. *Under the hypotheses of Proposition 5.3,* $\dim_{\mathrm{H}} \mathcal{P}_0 \le n-1$.

Proof. Step 1. Covering contraction. Fix $\delta > 0$ and put

$$s = n - 1 + \delta, \qquad d = n - 1 + \delta/2, \qquad a = \delta/(2N).$$

Choose $\theta < c_2$ so small that $C\theta^{\delta/2} < 1/4$, and then choose $\tau > 0$ so small that $C\theta^{-n}\tau^a < 1/4$. The constants here include those in the estimates below. With these θ, τ, take $h_* = h_*(\theta,\tau)$ and $\Theta_* = \Theta_*(\theta,\tau)$ from Proposition 5.3. Fix m and write

$$E_m = \{x \in \mathcal{P}_0 : \Theta_h(x) < \Theta_* \text{ for every } 0 < h < 1/m\} \quad \text{and} \quad \bar{h} = \min\{h_*, 1/m\}.$$

These sets cover $\mathcal{P}_0$. Fix $x \in E_m$ and $h \in (0, \bar{h})$.

For each E_m, we consider the following decompositions

$$\mathcal{A} = \{x' \in E_m \cap B_{r_h(x)}(x) : r_{\tau h}(x') \notin [c\theta r_h(x), C r_h(x)]\} \quad \text{and} \quad \mathcal{G} = (E_m \cap B_{r_h(x)}(x)) \setminus \mathcal{A}.$$

By Proposition 5.3 (ii), $\mathcal{A} \subset \mathcal{E}_{x,h}$. According to Proposition 5.3 (i)

$$\mathcal{A} \subset \bigcup_{i\in I} B_{\theta r_h(x)}(x_i), \quad \text{with } x_i \in \mathcal{A} \text{ and } \#I \le C\theta^{-(n-1)}.$$

For $\mathcal{G}$, choose a maximal $c\theta r_h(x)/2$-separated family of centers $\{x_j\}_{j\in J} \subset \mathcal{G}$. The balls of radius $c\theta r_h(x)/4$ about them are disjoint in $B_{2r_h(x)}(x)$, so there are at most $C\theta^{-n}$ centers. Since $r_{\tau h}(x_j) \ge c\theta r_h(x)$ by the definition of $\mathcal{G}$, maximality of the separated family then ensures

$$\mathcal{G} \subset \bigcup_{j\in J} B_{r_{\tau h}(x_j)}(x_j), \quad \text{with } x_j \in \mathcal{G},\ \#J \le C\theta^{-n} \text{ and } r_{\tau h}(x_j) \le C r_h(x).$$

Together these balls cover $E_m \cap B_{r_h(x)}(x)$. We also have

$$\begin{aligned} \sum_{i\in I} (\theta r_h(x))^d h^a + \sum_{j\in J} r_{\tau h}(x_j)^d (\tau h)^a &\le C\theta^{-(n-1)} (\theta r_h(x))^d h^a + C\theta^{-n} r_h(x)^d (\tau h)^a \\ &\le C\big(\theta^{\delta/2} + \theta^{-n}\tau^a\big) r_h(x)^d h^a \le \frac{1}{2} r_h(x)^d h^a. \end{aligned} \tag{5.17}$$

Step 2. Hausdorff content. Set

$$M = \sup_{\substack{x\in E_m \\ 0<h<\bar{h}}} \frac{\mathcal{H}^s_\infty(E_m \cap B_{r_h(x)}(x))}{r_h(x)^d h^a},$$

with $M = 0$ if E_m is empty. This supremum is finite, since (5.11) gives

$$\mathcal{H}^s_\infty(E_m \cap B_{r_h(x)}(x)) \le C_s r_h(x)^s \le C r_h(x)^d h^a.$$

All covering centers lie in E_m. For each $i \in I$, (5.10) gives $r_h(x_i) \ge c_2 r_h(x) > \theta r_h(x)$. Continuity in the height and $r_t(x_i) \to 0$ as $t \downarrow 0$ give a height $t \in (0, h]$ with $r_t(x_i) = \theta r_h(x)$. Since $t \le h < \bar{h}$, the definition of M gives

$$\mathcal{H}^s_\infty(E_m \cap B_{\theta r_h(x)}(x_i)) \le M(\theta r_h(x))^d t^a \le M(\theta r_h(x))^d h^a.$$

For each $j \in J$, the corresponding height is $\tau h < \bar{h}$, so

$$\mathcal{H}^s_\infty(E_m \cap B_{r_{\tau h}(x_j)}(x_j)) \le M r_{\tau h}(x_j)^d (\tau h)^a.$$

Apply subadditivity to the cover in Step 1 and use (5.17):

$$\begin{aligned}\mathcal{H}^s_\infty(E_m \cap B_{r_h(x)}(x)) &\le \sum_{i\in I} \mathcal{H}^s_\infty(E_m \cap B_{\theta r_h(x)}(x_i)) + \sum_{j\in J} \mathcal{H}^s_\infty(E_m \cap B_{r_{\tau h}(x_j)}(x_j)) \\ &\le M\left(\sum_{i\in I} (\theta r_h(x))^d h^a + \sum_{j\in J} r_{\tau h}(x_j)^d (\tau h)^a\right) \le \frac{M}{2} r_h(x)^d h^a.\end{aligned}$$

Dividing by $r_h(x)^d h^a$ and taking the supremum over $x \in E_m$ and $0 < h < \bar{h}$ gives $M \le M/2$, hence $M = 0$. A countable subcover of E_m by the balls in this supremum shows

$$\mathcal{H}^s_\infty(E_m) = 0.$$

Since zero Hausdorff content is equivalent to zero Hausdorff measure, $\mathcal{H}^s(E_m) = 0$. Taking the countable union over m and letting $\delta \downarrow 0$ completes the proof.

□

6. Higher dimensions

Let $n \ge 3$. We retain the localization and fixed sets $\mathcal{P}$ from Section 3. Lemma 5.2 is valid in every dimension. To use Propositions 5.3 and 5.4, we must establish the limit classification assumed in Proposition 5.3. We introduce an additional finite measure on the subgradient graph to estimate the transverse derivatives of the normalized boundary polynomial. Together with the target and source mass estimates, these estimates yield the alternatives in Proposition 6.2.

For the fixed potential v in (3.5), let v_1 be its Moreau regularization,

$$v_1(z) = \min_{y' \in \mathbb{R}^n} \{v(y') + \tfrac{1}{2}|z - y'|^2\}, \qquad z \in \mathbb{R}^n. \tag{6.1}$$

The minimizer $y(z)$ is unique. Take $x(z) = z - y(z)$. By the standard properties of Moreau regularization [21], $v_1 \in C^{1,1}$, both v_1 and $\frac{1}{2}|z|^2 - v_1$ are convex, and

$$Dv_1(z) = x(z). \tag{6.2}$$

The mapping $z \mapsto y(z)$ was introduced in [20], and can be written as

$$y(z) = z - Dv_1(z). \tag{6.3}$$

The maps x, y are Lipschitz continuous, $x(z) \in \partial v(y(z))$ and every pair $x \in \partial v(y)$ arises from exactly one $z = x + y$.

Consider $K(z) = D_z y$. Then $K = I - D^2 v_1$ almost everywhere. By (6.2) and (6.3)

$$K = K^{\mathsf{T}}, \qquad 0 \le K \le I, \qquad D_z x = I - K. \tag{6.4}$$

All graph integrals below are restricted to a fixed compact subset of the graph

$$\{(y, x) \in \mathbb{R}^n \times \mathbb{R}^n : x \in \partial v(y)\},$$

with pairs $(y, x) \in \Omega_0^* \times \Omega$, chosen to contain the sets used below. The margins in Definition 3.3 allow this choice uniformly at small heights.

6.1. **Weighted measures on the graph.**

Let $\Gamma_0 \Subset \Omega_0^* \times \Omega$ be a fixed compact graph subset and set

$$\mathcal{Z}_0 := \{x + y : (y, x) \in \Gamma_0\} = \{z \in \mathbb{R}^n : (y(z), x(z)) \in \Gamma_0\}.$$

Define $\widetilde{\nu}_\epsilon$ on the z-parameter space $\mathbb{R}^n$ by the density below on $\mathcal{Z}_0$, extended by zero outside $\mathcal{Z}_0$. Its pushforward $\widetilde{\mu}_\epsilon$ under $z \mapsto x(z)$ is a measure on the source space $\mathbb{R}^n$:

$$\begin{aligned} d\widetilde{\nu}_\epsilon(z) &= \mathbf{1}_{\{F(y(z)) \neq 0\}} |F(y(z))|^{-1+\epsilon} DF(y(z))^{\mathsf{T}} K(z) DF(y(z))\, dz \\ \widetilde{\mu}_\epsilon &= x_{\#} \widetilde{\nu}_\epsilon. \end{aligned} \tag{6.5}$$

The finiteness of these measures follows directly by integration by parts on the graph.

Lemma 6.1. *The measures $\widetilde{\nu}_\epsilon$ and $\widetilde{\mu}_\epsilon$ are finite.*

Proof. Choose a nonnegative $\chi \in C_c^1(y^{-1}(\Omega_0^*))$ equal to one on $\mathcal{Z}_0$. For $t > 0$, consider

$$b_t(s) = \int_0^s (|r| + t)^{-1+\epsilon}\, dr \quad \text{and} \quad |b_t(s)| \le |s|^\epsilon / \epsilon.$$

Since $K = Dy$ is symmetric, the Lipschitz chain rule gives, almost everywhere,

$$\begin{aligned} D_z[b_t(F(y))] &= (|F(y)| + t)^{-1+\epsilon} K DF(y), \\ \operatorname{div}_z(\chi DF(y)) &= D\chi \cdot DF(y) + \chi \operatorname{tr}(D^2 F(y) K). \end{aligned}$$

Integration by parts, using the compact support of χ, therefore gives

$$\begin{aligned} &\int_{\mathbb{R}^n} \chi \cdot (|F(y)| + t)^{-1+\epsilon} DF(y)^{\mathsf{T}} K DF(y)\, dz \\ &\qquad = -\int_{\mathbb{R}^n} b_t(F(y)) \big[D\chi \cdot DF(y) + \chi \operatorname{tr}(D^2 F(y) K)\big]\, dz \le C. \end{aligned}$$

The constant C is independent of t, by the bound for b_t, (6.4), and the bounds for $F, DF, D^2 F$ on the fixed compact set $y(\operatorname{supp} \chi)$. Restrict the left side to $\mathcal{Z}_0 \cap \{F \circ y \neq 0\}$ and let $t \downarrow 0$. Monotone convergence proves that $\widetilde{\nu}_\epsilon$ is finite. Its pushforward $\widetilde{\mu}_\epsilon$ is finite as well. □

6.2. **Regularized sections.**

For a pair (y_0, x_0) set $z_0 = x_0 + y_0$. According to (6.1), $v_1(z_0) = v(y_0) + \frac{1}{2}|x_0|^2$, and

$$v_1(z_0 + z) = \min_y \{v(y_0 + y) + \frac{1}{2}|x_0 + z - y|^2\}.$$

Completing the square then shows that

$$v_1(z_0 + z) - v_1(z_0) - x_0 \cdot z = \min_y \left\{v(y_0 + y) - v(y_0) - x_0 \cdot y + \tfrac{1}{2}|z - y|^2\right\}. \tag{6.6}$$

Together with $Dv_1(z_0) = x_0$, see (6.2), we conclude that

$$\tfrac{1}{2}(S_h[v](y_0, x_0) - y_0) + B_{\sqrt{h}} \subset S_h[v_1](z_0) - z_0 \subset S_h[v](y_0, x_0) - y_0 + B_{\sqrt{2h}}. \tag{6.7}$$

For the first inclusion, write $z = \xi/2 + q$ with $\xi \in S_h[v](y_0, x_0) - y_0$ and $|q| < \sqrt{h}$. Taking $y = \xi/2$ in (6.6) and using convexity shows

$$v_1(z_0 + z) - v_1(z_0) - x_0 \cdot z \le \frac{1}{2}(v(y_0 + \xi) - v(y_0) - x_0 \cdot \xi) + \frac{1}{2}|q|^2 < h.$$

On the other side, take $z \in S_h[v_1](z_0) - z_0$ and let y be the minimizer in (6.6). Then

$$v(y_0 + y) - v(y_0) - x_0 \cdot y \le h - \tfrac{1}{2}|z - y|^2.$$

Hence $y \in S_h[v](y_0, x_0) - y_0$. Since $v(y_0 + y) - v(y_0) - x_0 \cdot y \ge 0$, we obtain $|z - y| \le \sqrt{2h}$. The second inclusion follows.

Denote by $b_h + T_h B_1$ the inner normalizing ellipsoid of $S_h[v](y_0, x_0)$. Suppose $T_h = O_h \operatorname{diag}(a_i)$ for some orthogonal matrix O_h. Let

$$a_i' = \sqrt{a_i^2 + h}, \qquad \widehat{T}_h = O_h \operatorname{diag}(a_i'), \qquad \beta_h' = \sqrt{\beta(h)^2 + h}. \tag{6.8}$$

Then $(\det \widehat{T}_h)^2 = \prod_i (a_i^2 + h)$ and $\beta_h' \approx \beta(h)$ by (3.12). The product of the largest $n - 1$ semiaxes of the ellipsoid $x_0 + h\widehat{T}_h^{-\mathsf{T}} B_1$ is

$$C'_{\mathrm{cov}}(x_0, h) = \frac{\beta_h' h^{n-1}}{|\det \widehat{T}_h|}. \tag{6.9}$$

For $|Z| < 1$, take $z = b_h - y_0 + \widehat{T}_h Z$ and $y = b_h - y_0 + T_h Z \in S_h[v](y_0, x_0) - y_0$ in (6.6). Then

$$v_1(x_0 + b_h + \widehat{T}_h Z) - v_1(z_0) - x_0 \cdot (b_h - y_0 + \widehat{T}_h Z) \le h + \frac{1}{2}|(\widehat{T}_h - T_h)Z|^2 < \frac{3}{2}h, \tag{6.10}$$

since $\|\widehat{T}_h - T_h\| \le \sqrt{h}$ in the principal frame. In particular,

$$x_0 + b_h + \widehat{T}_h B_1 \subset S_{4h}[v_1](z_0).$$

With the fixed ρ from (4.26), let us set

$$G_h(x_0) = x_0 + b_h + \rho \widehat{T}_h B_1. \tag{6.11}$$

In the coordinates $z = x_0 + b_h + \widehat{T}_h Z$, the set $G_h(x_0)$ corresponds to B_ρ and the image of the section $S_{4h}[v_1](z_0)$ in these coordinates contains B_1. Thus $G_h(x_0)$ has distance at least $1 - \rho > 1/2$ from the section boundary in these coordinates, uniformly in x_0 and h.

To apply Lemma 4.1 to v_1, we use base point z_0, supporting slope x_0, height h, linear map $\widehat{T}_h$ and normalized center $Z_0 := \widehat{T}_h^{-1}(b_h - y_0)$. In the coordinates $z = z_0 + \widehat{T}_h Z$, the corresponding normalized potential is

$$\widetilde{v}_{1,h}(Z) = \frac{v_1(z_0 + \widehat{T}_h Z) - v_1(z_0) - x_0 \cdot \widehat{T}_h Z}{h}. \tag{6.12}$$

Since $\|\widehat{T}_h^{-1} T_h\| \le 1$ and $S_h[v](y_0, x_0) \subset b_h + nT_h B_1$, we have

$$|Z_0| \le |T_h^{-1}(b_h - y_0)| \le n.$$

Estimate (6.10) shows $\widetilde{v}_{1,h} \le 3/2$ on $B_1(Z_0)$. This verifies (4.6) for $\widetilde{v}_{1,h}$. Set $R := 2n + \sqrt{2}$. Since $\sqrt{h}\,\|\widehat{T}_h^{-1}\| \le 1$, we infer by the second conclusion in (6.7) that

$$\{\widetilde{v}_{1,h} < 1\} = \widehat{T}_h^{-1}(S_h[v_1](z_0) - z_0) \subset B_R.$$

For $|Z| \ge R$, take $t = R/|Z| \le 1$. Convexity gives $1 \le \widetilde{v}_{1,h}(tZ) \le t\widetilde{v}_{1,h}(Z)$. Hence

$$\widetilde{v}_{1,h}(Z) \ge |Z|/R - 1 \qquad (Z \in \mathbb{R}^n).$$

Consequently, for every $\xi \in \mathbb{R}^n$,

$$\widetilde{v}_{1,h}(Z_0 + \xi) + \widetilde{v}_{1,h}(Z_0 - \xi) \ge \frac{|Z_0 + \xi| + |Z_0 - \xi|}{R} - 2 \ge \frac{2|\xi|}{R} - 2.$$

This verifies (4.7) at the prescribed center Z_0 with $c = 2/(2n + \sqrt{2})$ and $C = 2$, uniformly in x_0 and h. Apply Lemma 4.1 to v_1 with $Q_i = G_{h_i}(x_i)$. Since $r_1 = 1$ and $\rho < 1/20$ by (4.26),

$$Q_i = z_i + \widehat{T}_i B_\rho(Z_i) \subset z_i + \widehat{T}_i B_{r_1/20}(Z_i),$$

where $z_i = x_i + Du(x_i)$ and $Z_i = \widehat{T}_i^{-1}(b_i - Du(x_i))$. In each height range $h/2 < h_i \le h$, the lemma selects a countable subfamily, indexed by j, such that

$$\sum_j \widetilde{\nu}_\epsilon(G_{h_j}(x_j)) \le C\widetilde{\nu}_\epsilon(\mathbb{R}^n) \quad \text{and} \quad \{x_i : h/2 < h_i \le h\} \subset \bigcup_j (x_j + Ch_j \widehat{T}_j^{-\mathsf{T}} B_1).$$

Let ν_0 be the measure defined in (4.2). Using (6.9) and (6.11), we replace (4.29) by

$$\Theta_h(x) = \frac{\nu_0(\Omega_h^*(x)) + \widetilde{\mu}_\epsilon(\Omega_h(x))}{h^\epsilon C_{\mathrm{cov}}(x,h)} + \frac{\widetilde{\nu}_\epsilon(G_h(x))}{h^\epsilon C'_{\mathrm{cov}}(x,h)}. \tag{6.13}$$

Define $\mathcal{P}_+$ and $\mathcal{P}_0$ by (4.30) using this function. Since $a_i' \ge \sqrt{h}$, each semiaxis of $h\widehat{T}_h^{-\mathsf{T}} B_1$ is at most $\sqrt{h}$. The finiteness in Lemma 6.1, the selection just described, and Lemma 4.2 therefore yield the same covering estimate for the third term. The proof of Proposition 4.3, now with three alternatives instead of two, shows

$$\dim_{\mathrm{H}} \mathcal{P}_+ \le n - 1 + 2\epsilon.$$

6.3. **Limits of normalized sections.**

Let $x_j \in \mathcal{P}$, $y_j = Du(x_j)$ and $h_j \downarrow 0$. Write $C_{\mathrm{cov},j} = C_{\mathrm{cov}}(x_j, h_j)$ and $C'_{\mathrm{cov},j} = C'_{\mathrm{cov}}(x_j, h_j)$, with the covering costs defined in (4.28) and (6.9), respectively. Let $b_j + T_j B_1$ be a normalizing ellipsoid of $S_{h_j}[v](y_j, x_j)$ as in (2.4), with ordered semiaxis lengths $a_{1,j} \ge \cdots \ge a_{n,j} > 0$.

Recall that $\beta_j = \|T_j\| = a_{1,j}$. Denote by $\tilde{v}_j$ the normalized target potential and by $\tilde{u}_j$ its Legendre transform defined in (5.1). For all sufficiently large j, (4.1) and (3.5) allow us to use the source normalization (4.22) with $v^* = u$ for $X \in B_{2\rho}$. Let $F_j(Y) = F(y_j + T_j Y)$. Define

$$\sigma_{i,j} = a_{i,j}^2/h_j \quad \text{and} \quad \kappa_j = (\det T_j)^2/h_j^n = \prod_{i=1}^n \sigma_{i,j}. \tag{6.14}$$

Thus κ_j is the scaling factor for the Monge-Ampère density under the normalization (5.1). Choose orthonormal bases in the domain and range of T_j in which T_j is diagonal, and pass to a subsequence on which these bases converge. After passing to a further subsequence, $\tilde{u}_j \to \tilde{u}$ locally uniformly on $B_{2\rho}$, by the compactness argument in Step 1 of the proof of Proposition 5.1. By Lemma 3.5, after passing to a further subsequence, the convergence (5.3) holds for a nonzero polynomial P of degree at most d_0.

Proposition 6.2. *Suppose $\Theta_{h_j}(x_j) \to 0$, with Θ_h given by* (6.13)*. After a subsequence,*

$$T_j/\beta_j \to e^* \otimes e \quad \textit{and} \quad |e^*| = |e| = 1.$$

At least one of the following holds:

1. *$P(Y) = p(e \cdot Y)$ for a nonzero polynomial p of degree at most d_0.*
2. *$\#\{D_e\tilde{u}(X) : X \in B_\rho,\ \tilde{u}$ is differentiable at $X\} \le d_0$.*

Proof. Write $\Theta_j = \Theta_{h_j}(x_j)$. We work in the converging orthonormal bases chosen above.

Step 1. Source estimate. Estimates (3.11), (3.12) and (3.15) show

$$\kappa_j \ge c, \quad a_{n,j} \ge c h_j \quad \text{and} \quad c h_j \le \gamma_j \le C\beta_j. \tag{6.15}$$

By (4.28), the estimate (5.4), valid in all dimensions, and (6.14), we obtain

$$\frac{\nu_0(\Omega^*_{h_j}(x_j))}{h_j^\epsilon C_{\mathrm{cov},j}} \ge c \left(\frac{\gamma_j}{h_j}\right)^\epsilon \frac{\kappa_j h_j}{\beta_j \gamma_j}. \tag{6.16}$$

By (6.15), the right-hand side is at least $c h_j/\beta_j^2 = c/\sigma_{1,j}$. As $\Theta_{h_j}(x_j) \to 0$, we find that

$$\sigma_{1,j} \to \infty. \tag{6.17}$$

Almost every $x \in \Omega_{h_j}(x_j)$ belongs to the regular set Ω' of Lemma 2.1. At such x, we have $D^2u(x) = [D^2v(y)]^{-1}$ with $y = Du(x)$. By (6.2) and (6.3), $z = y + Dv(y)$. Differentiating this identity and using $K = D_z y$ as in (6.4), we obtain

$$K = (I + D^2v(y))^{-1} \quad \text{and} \quad dz = \det(I + D^2v(y))\, dy.$$

Changing variables in (6.5) and using (3.7), we obtain

$$\widetilde{\mu}_\epsilon(\Omega_{h_j}(x_j)) \ge \lambda/\Lambda \int_{\Omega_{h_j}(x_j)} F(Du)^{-1+\epsilon} DF(Du)^{\mathsf{T}} \operatorname{cof}(I + D^2v(y)) DF(Du)\, dx, \tag{6.18}$$

where $\operatorname{cof} A = (\det A)A^{-1}$ denotes the cofactor matrix. For $M \ge 1$, let

$$\mathcal{Q}_j(M) = \{X \in B_\rho : D^2\tilde{u}_j(X) \le MI\}.$$

Choose $\chi \in C_c^\infty(B_{2\rho})$ with $\chi \geq 0$ and $\chi = 1$ on B_ρ. The uniform bounds for $\tilde{u}_j$ imply

$$\int_{B_\rho} \operatorname{tr} D^2\tilde{u}_j \, dX \leq \int \chi \, d(\Delta\tilde{u}_j) = \int \tilde{u}_j \Delta\chi \, dX \leq C.$$

It follows that

$$|B_\rho \setminus \mathcal{Q}_j(M)| \leq C/M. \tag{6.19}$$

For all sufficiently large j and almost every $X \in \mathcal{Q}_j(M)$, write

$$x = x_j + h_j T_j^{-\mathsf{T}} X. \tag{6.20}$$

Then $D^2u(x) \leq Mh_j^{-1} T_j T_j^{\mathsf{T}}$. Thus, for $y = Du(x)$ at such x,

$$D^2v(y) \geq M^{-1} h_j (T_j T_j^{\mathsf{T}})^{-1}.$$

Therefore, in the principal-axis coordinates of the target ellipsoid,

$$\operatorname{cof}(I + D^2v(y)) \geq c_M \operatorname{diag}\Big(\prod_{i\neq l}(1+\sigma_{i,j}^{-1})\Big)_{l=1}^n. \tag{6.21}$$

At these points,

$$0 < F(Du(x)) \leq C\gamma_j \quad \text{and} \quad dx = h_j^n |\det T_j|^{-1} \, dX. \tag{6.22}$$

Recall that $P_j(Y) = \gamma_j^{-1} F(y_j + T_j Y)$ in (5.3). Differentiating it and using (5.5) and (6.20),

$$[DF(Du(x))]_l = \frac{\gamma_j}{a_{l,j}} \partial_l P_j(D\tilde{u}_j(X)). \tag{6.23}$$

Restricting the integral in (6.18) to the points corresponding to $\mathcal{Q}_j(M)$ and using (6.21), (6.22) and (6.23), we obtain

$$\begin{aligned}
\widetilde{\mu}_\epsilon(\Omega_{h_j}(x_j)) &\geq c_M \gamma_j^{1+\epsilon} \frac{h_j^n}{|\det T_j|} \int_{\mathcal{Q}_j(M)} \sum_{l=1}^n a_{l,j}^{-2} |\partial_l P_j(D\tilde{u}_j(X))|^2 \prod_{i\neq l}(1+\sigma_{i,j}^{-1}) \, dX \\
&= c_M \gamma_j^{1+\epsilon} \frac{C_{\mathrm{cov},j}}{\kappa_j \beta_j} \int_{\mathcal{Q}_j(M)} \sum_{l=1}^n |\partial_l P_j(D\tilde{u}_j(X))|^2 \prod_{i\neq l}(1+\sigma_{i,j}) \, dX.
\end{aligned}$$

Dividing by $h_j^\epsilon C_{\mathrm{cov},j}$ and multiplying by (6.16), we use $\gamma_j/h_j \geq c$ to obtain

$$\int_{\mathcal{Q}_j(M)} \sum_{l=1}^n \frac{\prod_{i\neq l}(1+\sigma_{i,j})}{\sigma_{1,j}} |\partial_l P_j(D\tilde{u}_j)|^2 \, dX \leq C_M \Theta_j^2. \tag{6.24}$$

Step 2. Boundary polynomial. Fix an index i_* with $a_{i_*,j} \approx \beta_j$. For $l \neq i_*$, the coefficient of $|\partial_l P_j(D\tilde{u}_j)|^2$ in (6.24) is bounded from below by a positive constant. It then follows from (6.24) and (6.19) that

$$\int_{B_\rho} |\partial_l P_j(D\tilde{u}_j)|^2 \, dX \leq C_M \Theta_j^2 + C/M.$$

Sending $j \to \infty$, $M \to \infty$ and using $D\tilde{u}_j \to D\tilde{u}$ almost everywhere, we obtain

$$\partial_l P(D\tilde{u}(X)) = 0 \quad \text{for a.e. } X \in B_\rho \quad (l \neq i_*,\ a_{i_*,j} \approx \beta_j). \tag{6.25}$$

Since $\sigma_{l,j} \geq \sigma_{n,j}$ and $\sigma \mapsto \sigma/(1+\sigma)$ is increasing on $(0,\infty)$,

$$a_{l,j}^2 \prod_{i\neq l}(1+\sigma_{i,j}) = \frac{h_j\sigma_{l,j}}{1+\sigma_{l,j}} \prod_{i=1}^{n}(1+\sigma_{i,j}) \geq \frac{h_j\sigma_{n,j}}{1+\sigma_{n,j}} \prod_{i=1}^{n}(1+\sigma_{i,j}) = a_{n,j}^2 \prod_{i<n}(1+\sigma_{i,j}).$$

Since $|DF| \geq 1$, we conclude by (6.23) that

$$\sum_{l=1}^{n} \frac{\prod_{i\neq l}(1+\sigma_{i,j})}{\sigma_{1,j}} |\partial_l P_j(D\tilde{u}_j)|^2 \geq \frac{a_{n,j}^2}{\gamma_j^2\sigma_{1,j}} \prod_{i<n}(1+\sigma_{i,j}) \geq \frac{a_{n,j}^2 a_{2,j}^2}{h_j\gamma_j^2}. \tag{6.26}$$

Here the last inequality uses $n \geq 3$ to retain both $\sigma_{1,j}$ and $\sigma_{2,j}$ in the product. By (6.19), fix M with $|\mathcal{Q}_j(M)| \geq |B_\rho|/2$. Since $a_{n,j} \geq ch_j$, (6.24) and (6.26) imply

$$\frac{a_{2,j}\sqrt{h_j}}{\gamma_j} \leq C\Theta_j \to 0. \tag{6.27}$$

By (6.17) and the ordering of the $\sigma_{i,j}$, after passing to a subsequence there is $m \in \{1,\dots,n\}$,

$$\sigma_{i,j} \to \infty \text{ for } i \leq m \quad \text{and} \quad \sigma_{i,j} \text{ is bounded for } i > m.$$

For $k > m$ and $l \geq 2$, we have $a_{k,j} \leq C\sqrt{h_j}$ and $a_{l,j} \leq a_{2,j}$. The uniform C^2 bound for F and (6.27) therefore imply, on bounded sets,

$$|\partial_l\partial_k P_j| \leq C\frac{a_{k,j}a_{l,j}}{\gamma_j} \leq C\frac{a_{2,j}\sqrt{h_j}}{\gamma_j} \to 0 \quad (k > m,\ l \geq 2). \tag{6.28}$$

Thus $\partial_k P$ depends only on Y_1 for each $k > m$, and

$$P(Y) = p_0(Y_1,\dots,Y_m) + \sum_{k=m+1}^{n} p_k(Y_1)Y_k. \tag{6.29}$$

Here $p_0(Y_1,\dots,Y_m) = P(Y_1,\dots,Y_m,0,\dots,0)$ is a polynomial of degree at most d_0 with $p_0(0) = 0$; and p_k is the polynomial defined by $\partial_k P(Y) = p_k(Y_1)$, of degree at most $d_0 - 1$.

Step 3. Graph estimate. Let b_j be the target ellipsoid center and take $\widehat{T}_j$ from (6.8) at h_j. Set

$$z = x_j + b_j + \widehat{T}_j Z \quad \text{and} \quad Y_j(Z) = T_j^{-1}(y(z) - y_j), \quad (Z \in B_\rho) \tag{6.30}$$

and

$$\widehat{P}_j(Z) = P_j(Y_j(Z)), \quad (Z \in B_\rho).$$

Applying (6.6) with $(y_0, x_0) = (y_j, x_j)$ and evaluating at the minimizer, we have

$$\begin{aligned} v_1(z) - v_1(x_j + y_j) - x_j \cdot (z - x_j - y_j) &= v(y(z)) - v(y_j) - x_j \cdot (y(z) - y_j) + \tfrac{1}{2}|x(z) - x_j|^2 \\ &= h_j\tilde{v}_j(Y_j(Z)) + \tfrac{1}{2}|x(z) - x_j|^2. \end{aligned} \tag{6.31}$$

Here $Y_j(Z)$ is defined in (6.30). The two terms on the right are nonnegative, and by (6.10) and (6.30), each term is bounded by $3h_j/2$. Therefore

$$|x(z) - x_j| \leq C\sqrt{h_j} \quad \text{and} \quad \tilde{v}_j(Y_j(Z)) \leq 3/2. \tag{6.32}$$

Estimate (4.23) then bounds $Y_j(Z)$ uniformly, thus $|\widehat{P}_j(Z)| \le C$ on B_ρ. By (6.4),

$$|D_z(F\circ y)|^2 = |KDF(y)|^2 \le DF(y)^{\mathsf{T}} KDF(y).$$

The derivative on the left is zero almost everywhere on $\{F\circ y = 0\}$. On its complement, $|F\circ y| \le C\gamma_j$. Consequently (6.5) controls the integral over the whole ball:

$$\widetilde{\nu}_\epsilon(G_{h_j}(x_j)) \ge c\gamma_j^{-1+\epsilon}\int_{G_{h_j}(x_j)} |D_z(F\circ y)|^2\,dz = c\gamma_j^{1+\epsilon}|\det \widehat{T}_j| \int_{B_\rho}\sum_{l=1}^n \frac{|\partial_l \widehat{P}_j|^2}{(a'_{l,j})^2}\,dZ.$$

Dividing by $h_j^\epsilon C'_{\mathrm{cov},j}$ and multiplying by (6.16), we obtain

$$\begin{aligned}\frac{\widetilde{\nu}_\epsilon(G_{h_j}(x_j))}{h_j^\epsilon C'_{\mathrm{cov},j}}\frac{\nu_0(\Omega^*_{h_j}(x_j))}{h_j^\epsilon C_{\mathrm{cov},j}} &\ge c\left(\frac{\gamma_j}{h_j}\right)^{2\epsilon}\frac{\kappa_j h_j|\det \widehat{T}_j|}{\beta_j C'_{\mathrm{cov},j}}\int_{B_\rho}\sum_{l=1}^n\frac{|\partial_l\widehat{P}_j|^2}{(a'_{l,j})^2}\,dZ\\ &= c\left(\frac{\gamma_j}{h_j}\right)^{2\epsilon}\frac{\kappa_j h_j}{\beta_j\beta'_j}\int_{B_\rho}\sum_{l=1}^n\prod_{i\ne l}(1+\sigma_{i,j})|\partial_l\widehat{P}_j|^2\,dZ, \qquad (6.33)\end{aligned}$$

where we used $C'_{\mathrm{cov},j} = \beta'_j h_j^{n-1}/|\det \widehat{T}_j|$ and $(a'_{l,j})^2 = h_j(1+\sigma_{l,j})$. By (6.17)

$$\beta'_j/\beta_j = \sqrt{1+\sigma_{1,j}^{-1}} \to 1.$$

This together with $\frac{h_j}{\beta_j\beta'_j} = \frac{1}{\sigma_{1,j}}\frac{\beta_j}{\beta'_j}$ implies the coefficient of $|\partial_l\widehat{P}_j|^2$ is, up to fixed factors,

$$\left(\frac{\gamma_j}{h_j}\right)^{2\epsilon}\frac{\kappa_j}{\sigma_{1,j}}\prod_{i\ne l}(1+\sigma_{i,j}).$$

Using $\kappa_j \ge c$ and $\gamma_j/h_j \ge c$ from (6.15), together with (6.33) and (6.13), we obtain

$$\int_{B_\rho}\sum_{l=1}^n \frac{\prod_{i\ne l}(1+\sigma_{i,j})}{\sigma_{1,j}}|\partial_l\widehat{P}_j|^2\,dZ \le C\Theta_j^2. \qquad (6.34)$$

In particular, $\partial_l\widehat{P}_j \to 0$ in L^2 whenever $l \ne i_*$ and $a_{i_*,j} \approx \beta_j$.

Let $Y_j^0 = T_j^{-1}(b_j - y_j) \to Y^0$ after a subsequence. We infer by (6.30)

$$[Y_j(Z)]_i = [Y_j^0]_i + \frac{a'_{i,j}}{a_{i,j}}Z_i - \frac{[x(z)-x_j]_i}{a_{i,j}} \to [Y^0]_i + Z_i \quad \text{for } i=1,\dots,m, \qquad (6.35)$$

uniformly, using the displacement bound in (6.32), $a'_{i,j}/a_{i,j}\to 1$, and $\sqrt{h_j}/a_{i,j}\to 0$. If all p_k with $k>m$ vanish in (6.29), the coordinate convergence (6.35), the uniform boundedness of Y_j and (5.3) imply that $\widehat{P}_j \to P(Y^0+Z)$ uniformly on B_ρ. Passing to weak derivatives in (6.34) then yields

$$\partial_l P \equiv 0 \quad (l\ne i_*,\ a_{i_*,j}\approx\beta_j), \qquad (6.36)$$

provided that $p_k \equiv 0$ for every $k>m$ in (6.29). Unlike (6.25), this identity is global but requires the extra condition above.

Step 4. Classification. Suppose $a_{2,j} \geq c\beta_j$ along a subsequence. Then, by (6.27),

$$\beta_j \sqrt{h_j}/\gamma_j \to 0.$$

For $k > m$, the argument for (6.28) therefore also applies with $l = 1$. As a result,

$$p_k(Y_1) = \partial_k P(Y) \equiv \text{const.}, \qquad k = m+1, \dots, n.$$

Applying (6.25) with $i_* = 1$ makes $p_k \equiv 0$. Since now $i_* = 1, 2$, we further deduce by (6.36) that $DP(Y) \equiv 0$. This contradicts $P(0) = 0$ and $P \not\equiv 0$. Hence

$$\frac{a_{2,j}}{\beta_j} \to 0.$$

As in Step 2 of the proof of Proposition 5.1, since $a_{i,j} \leq a_{2,j}$ for $i \geq 2$, we have $T_j/\beta_j \to e^* \otimes e$, with $|e| = |e^*| = 1$.

If some p_k with $k > m$ is nonzero, then (6.25) and (6.29) imply $p_k(e \cdot D\tilde{u}) = 0$ almost everywhere. By continuity of the gradient on the differentiability set of a convex function, this identity holds at every differentiability point. The real roots of p_k give alternative (2).

Otherwise all these p_k vanish. By virtue of (6.36) with $i_* = 1$, we find $\partial_l P \equiv 0$ for every $l \geq 2$. As in Step 3 of the proof of Proposition 5.1, this yields $P(Y) = p(e \cdot Y)$, where $p(t) = P(te)$ is nonzero and has degree at most d_0, proving alternative (1). □

Proof of Proposition 1.2. Fix $\delta > 0$. Take $0 < \alpha < \min\{1, \delta\}$, $0 < \epsilon < \min\{1, \delta/2\}$ and $N = 2/\alpha$. By Lemma 3.1, $\dim_{\mathrm{H}}(E \cap Z_\alpha) \leq n - 1 + \alpha < n - 1 + \delta$. By Lemmas 3.1 and 3.2, the remaining set $E \setminus Z_\alpha$ admits a countable cover by sets $\mathcal{P}$ as in Definition 3.3. The section and Taylor estimates in Lemmas 3.4 and 3.5 are uniform on each such set.

On each subset, we split $\mathcal{P} = \mathcal{P}_+ \cup \mathcal{P}_0$. The selection argument of Proposition 4.3 shows $\dim_{\mathrm{H}} \mathcal{P}_+ \leq n - 1 + 2\epsilon < n - 1 + \delta$. Proposition 5.1 for $n = 2$ and Proposition 6.2 for $n \geq 3$ verify the hypotheses of Proposition 5.3. Proposition 5.4 therefore gives $\dim_{\mathrm{H}} \mathcal{P}_0 \leq n - 1$. Countable stability and then $\delta \downarrow 0$ yield $\dim_{\mathrm{H}} E \leq n - 1$. □

Proof of Theorem 1.1. By (2.2) and Lemma 2.1, $\Sigma = A \cup E$ and $\dim_{\mathrm{H}} A \leq n - 1$. Proposition 1.2 gives $\dim_{\mathrm{H}} E \leq n - 1$. The assertion follows. □

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References

[1] S. Arya, G. D. da Fonseca, and D. M. Mount, *Optimal area-sensitive bounds for polytope approximation*, Discrete Comput. Geom. **75** (2026), no. 3, 709–746. doi:10.1007/s00454-025-00815-5.

[2] I. Bárány and D. G. Larman, *Convex bodies, economic cap coverings, random polytopes*, Mathematika **35** (1988), no. 2, 274–291. doi:10.1112/S0025579300015266.

[3] Y. Brenier, *Décomposition polaire et réarrangement monotone des champs de vecteurs*, C. R. Acad. Sci. Paris Sér. I Math. **305** (1987), no. 19, 805–808.

[4] Y. Brenier, *Polar factorization and monotone rearrangement of vector-valued functions*, Comm. Pure Appl. Math. **44** (1991), no. 4, 375–417. doi:10.1002/cpa.3160440402.

[5] H. Brönnimann, B. Chazelle, and J. Pach, *How hard is half-space range searching?*, Discrete Comput. Geom. **10** (1993), 143–155. doi:10.1007/BF02573971.

[6] L. A. Caffarelli, *A localization property of viscosity solutions to the Monge-Ampère equation and their strict convexity*, Ann. of Math. (2) **131** (1990), no. 1, 129–134. doi:10.2307/1971509.

[7] L. A. Caffarelli, *Some regularity properties of solutions of Monge Ampère equation*, Comm. Pure Appl. Math. **44** (1991), no. 8–9, 965–969. doi:10.1002/cpa.3160440809.

[8] L. A. Caffarelli, *The regularity of mappings with a convex potential*, J. Amer. Math. Soc. **5** (1992), no. 1, 99–104. doi:10.1090/S0894-0347-1992-1124980-8.

[9] S. Chen and J. Liu, *Regularity of singular set in optimal transportation*, Invent. Math. **242** (2025), no. 1, 1–44. doi:10.1007/s00222-025-01353-w.

[10] S. Chen, Y. Li, and J. Liu, *Optimal (partial) transport to non-convex polygonal domains*, preprint, arXiv:2311.15655v4, 2026. doi:10.48550/arXiv.2311.15655.

[11] S. Chen, Q.-R. Li, Y. Li, J. Liu, and X. Wu, *Optimal transport to non-convex polyhedral domains*, preprint, arXiv:2610.18756, 2026.

[12] G. De Philippis and A. Figalli, *The Monge–Ampère equation and its link to optimal transportation*, Bull. Amer. Math. Soc. (N.S.) **51** (2014), no. 4, 527–580. doi:10.1090/S0273-0979-2014-01459-4.

[13] G. De Philippis and A. Figalli, *Partial regularity for optimal transport maps*, Publ. Math. Inst. Hautes Études Sci. **121** (2015), 81–112. doi:10.1007/s10240-014-0064-7.

[14] A. Figalli, *Regularity properties of optimal maps between nonconvex domains in the plane*, Comm. Partial Differential Equations **35** (2010), no. 3, 465–479. doi:10.1080/03605300903307673.

[15] A. Figalli, *The Monge-Ampère Equation and Its Applications*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, Zürich, 2017. doi:10.4171/170.

[16] A. Figalli and Y.-H. Kim, *Partial regularity of Brenier solutions of the Monge-Ampère equation*, Discrete Contin. Dyn. Syst. **28** (2010), no. 2, 559–565. doi:10.3934/dcds.2010.28.559.

[17] F. John, *Extremum problems with inequalities as subsidiary conditions*, in *Studies and Essays Presented to R. Courant on his 60th Birthday*, Interscience, New York, 1948, 187–204.

[18] J. Kitagawa and R. J. McCann, *Free discontinuities in optimal transport*, Arch. Ration. Mech. Anal. **232** (2019), no. 3, 1505–1541. doi:10.1007/s00205-018-01348-3.

[19] A. M. Macbeath, *A theorem on non-homogeneous lattices*, Ann. of Math. (2) **56** (1952), no. 2, 269–293. doi:10.2307/1969800.

[20] J.-J. Moreau, *Fonctions convexes duales et points proximaux dans un espace hilbertien*, C. R. Acad. Sci. Paris **255** (1962), 2897–2899.

[21] J.-J. Moreau, *Proximité et dualité dans un espace hilbertien*, Bull. Soc. Math. France **93** (1965), 273–299. doi:10.24033/bsmf.1625.

[22] C. Mooney, *Partial regularity for singular solutions to the Monge-Ampère equation*, Comm. Pure Appl. Math. **68** (2015), no. 6, 1066–1084. doi:10.1002/cpa.21534.

[23] L. Simon, *Introduction to Geometric Measure Theory*, lecture notes, Stanford University, 2014. Author's online text.

[24] Y. Yu, *Singular set of a convex potential in two dimensions*, Comm. Partial Differential Equations **32** (2007), no. 10–12, 1883–1894. doi:10.1080/03605300701318757.

[25] L. Zajíček, *On the differentiation of convex functions in finite and infinite dimensional spaces*, Czechoslovak Math. J. **29** (104) (1979), no. 3, 340–348. doi:10.21136/CMJ.1979.101616.

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